Session 1813

Networking Tutorials

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Objectives

- Provide some basic understanding of queuing phenomena
- Explain the available solution approaches and associated trade-offs
- Give guidelines on ho ns and solutions

Outline

- Basic concepts
- Source models
- Service models (demo)
- Single-queue systems
-
- Networks
- Hybrid simulation (demo)

Outline

- **Basic concepts**
  - **Performance measures**
  - **Solution methodologies**
  - **Queuing system concepts**
  - **Stability and steady-st**
-
- Service models (demo)
- Single-queue systems
- Priority/shared service systems
- Networks of queues
- Hybrid simulation (demo)

Performance Measures

- Delay
- Delay variation (jitter)
- Packet loss
- Efficient sharing of ba
-

  file transfe
- Challenge: Provide adequate performance for (possibly) heterogeneous traffic

Solution Methodologies

- Analytical results (formulas)
  - Pros: Quick answers, insight
  - Cons: Often inaccurate or inapplicable
- Explicit simulation
- Hybrid si
  - Intermediate solution approach
  - Combines advantages and disadvantages of analysis and simulation

Examples of Applications

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Queuing System Concepts: Arrival Rate, Occupancy, Time in the System

- Queuing system
  - Data network where packets arrive, wait in various queues, receive service at various points, and exit after some time
- Arrival rate
- Occupanc
  - Number of packets in the system (averaged over a long time)
- Time in the system (delay)
  - Time from packet entry to exit (averaged over many packets)

Stability and Steady-State

- A single queue system is stable if

  `packet arrival rate < system transmission capacity`

- For a single queue, the ratio

  `packet arrival ra                    sion capacity`

  i

- In an unsta eues and/or get dropped
- For unstable systems with large buffers some packet delays become very large
  - Flow/admission control may be used to limit the packet arrival rate
  - Prioritization of flows keeps delays bounded for the important traffic
- Stable systems with time-stationary arrival traffic approach a steady-state

Little's Law

- For a given arrival rate, the time in the system is proportional to packet occupancy

  N = ✦ T

  where

  ✦

  T: average delay (time in the system) per packet

- Examples:
  - On rainy days, streets and highways are more crowded
  - Fast food restaurants need a smaller dining room than regular restaurants with the same customer arrival rate
  - Large buffering together with large arrival rate cause large delays

Explanation of Little's Law

- Amusement park analogy: people arrive, spend time at various sites, and leave
- They pay \$1 per unit time in the park
- The rate at which the p unit time (N: a
-

traffic arri
- Over a long horizon:

Rate of park earnings = Rate of people's payment

or

$N = ✦ T$

Delay is Caused by Packet Interference

- If arrivals are regular or sufficiently spaced apart, no queuing delay occurs

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Irregular but Spaced Apart Traffic

Burstiness Causes Interference

- Note that the departures are less bursty

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Burstiness Example

Different Burstiness Levels at Same Packet Rate

Source: Fei Xue and S. J. Ben Yoo, UCDavis, "**On the Generation and Shaping Self-similar Traffic in Optical Packet-switched Networks", OPNETWORK 2002**

Packet Length Variation Causes Interference

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Regular arrivals, irregular packet lengths

High Utilization Exacerbates Interference

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As the work arrival rate:

(packet arrival rate * packet length)

increases, the opportunity for interference increases

Bottlenecks

- Types of bottlenecks
  - At access points (flow control, prioritization, QoS enforcement needed)
  - At points within the network core
  - Isolated (can be analyzed in isolation)

-
  - High loa
  - Convergence of sufficient number of moderate load sessions at the same queue

Bottlenecks Cause Shaping

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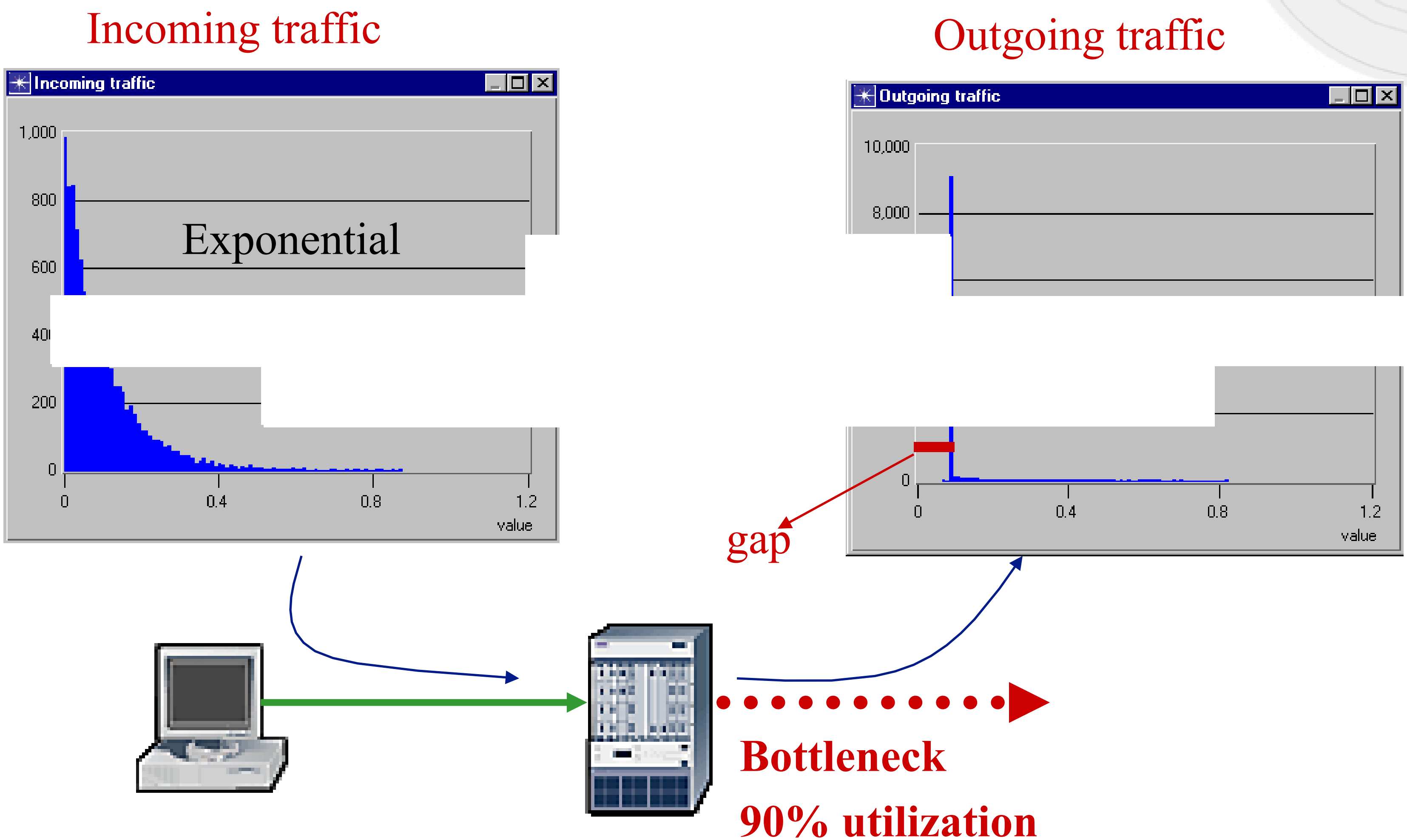
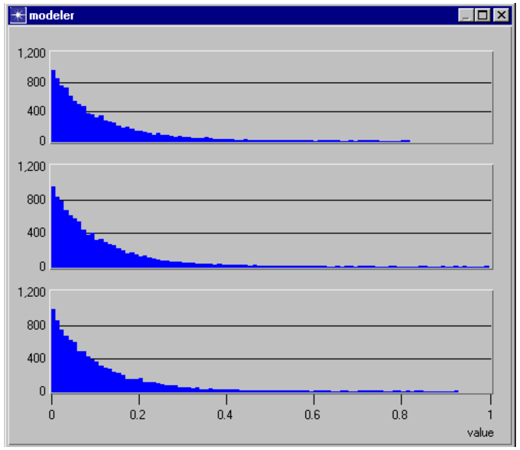
- The departure traffic from a bottleneck is more regular than the arrival traffic
- The inter-departure time between two packets is at least as large as the transmission time of the 2nd packet

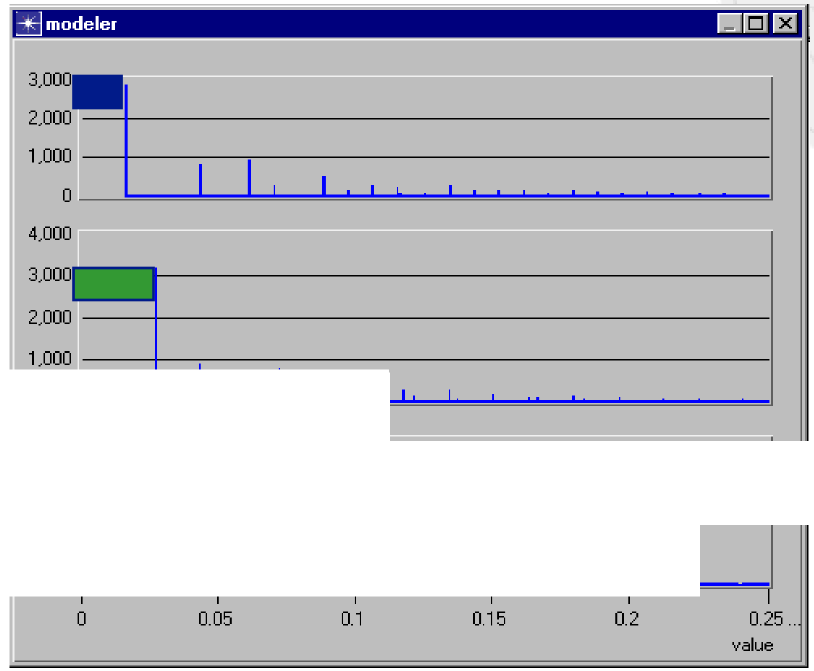
Bottlenecks Cause Shaping

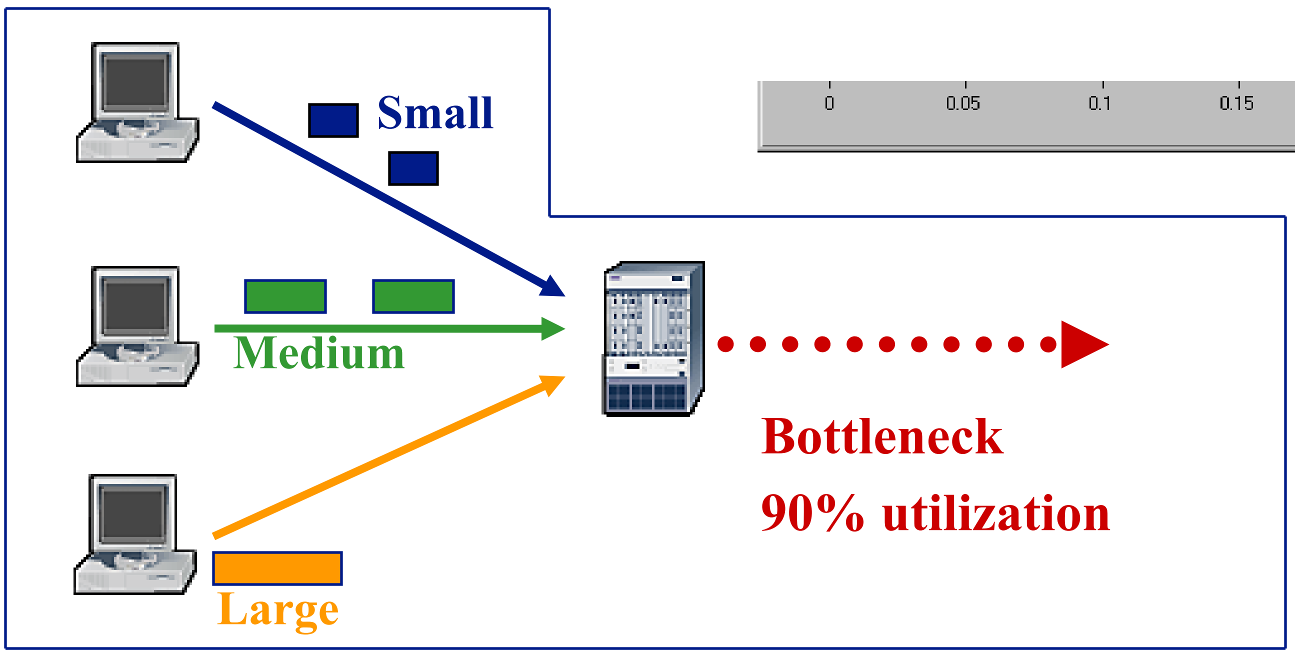


Incoming traffic



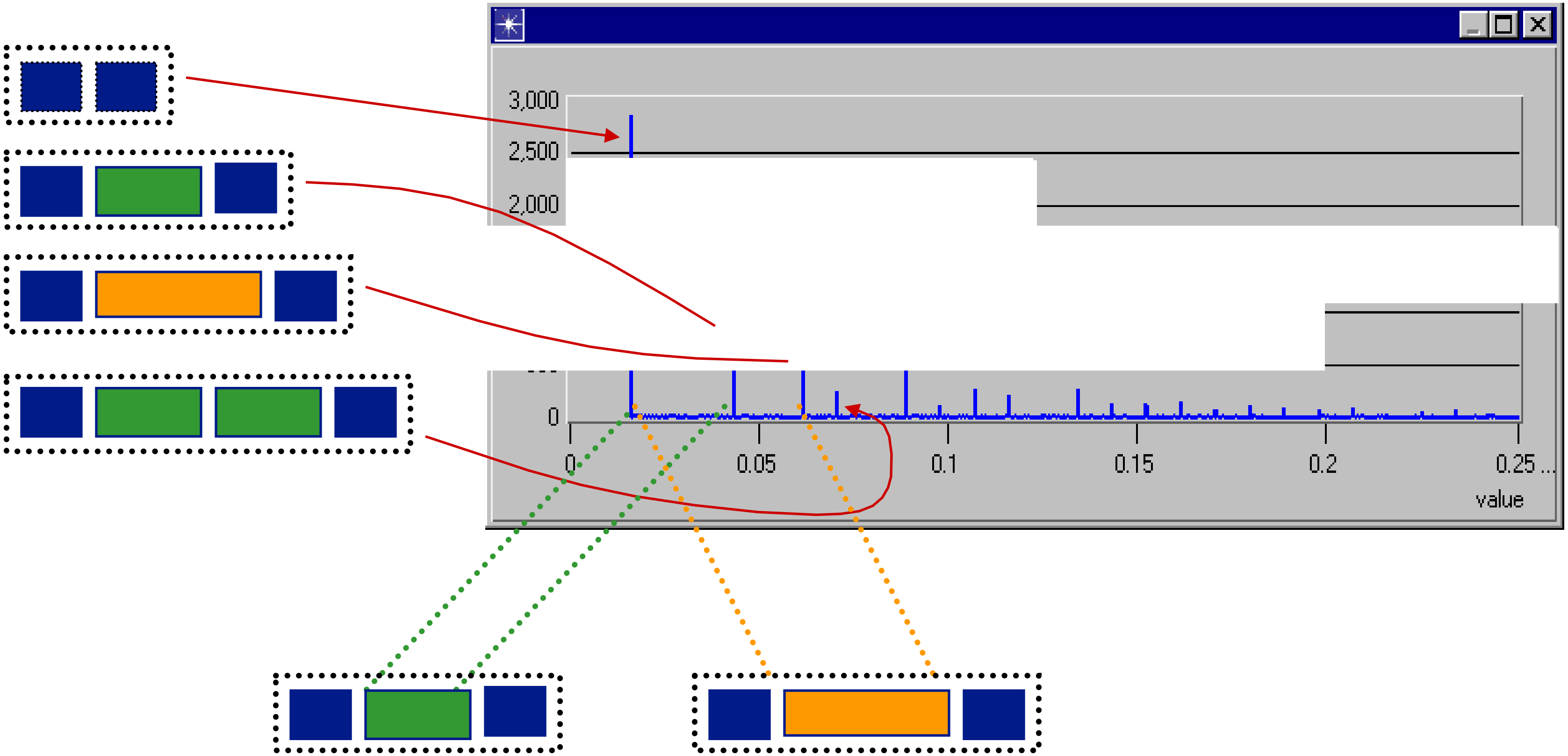
Outgoing traffic





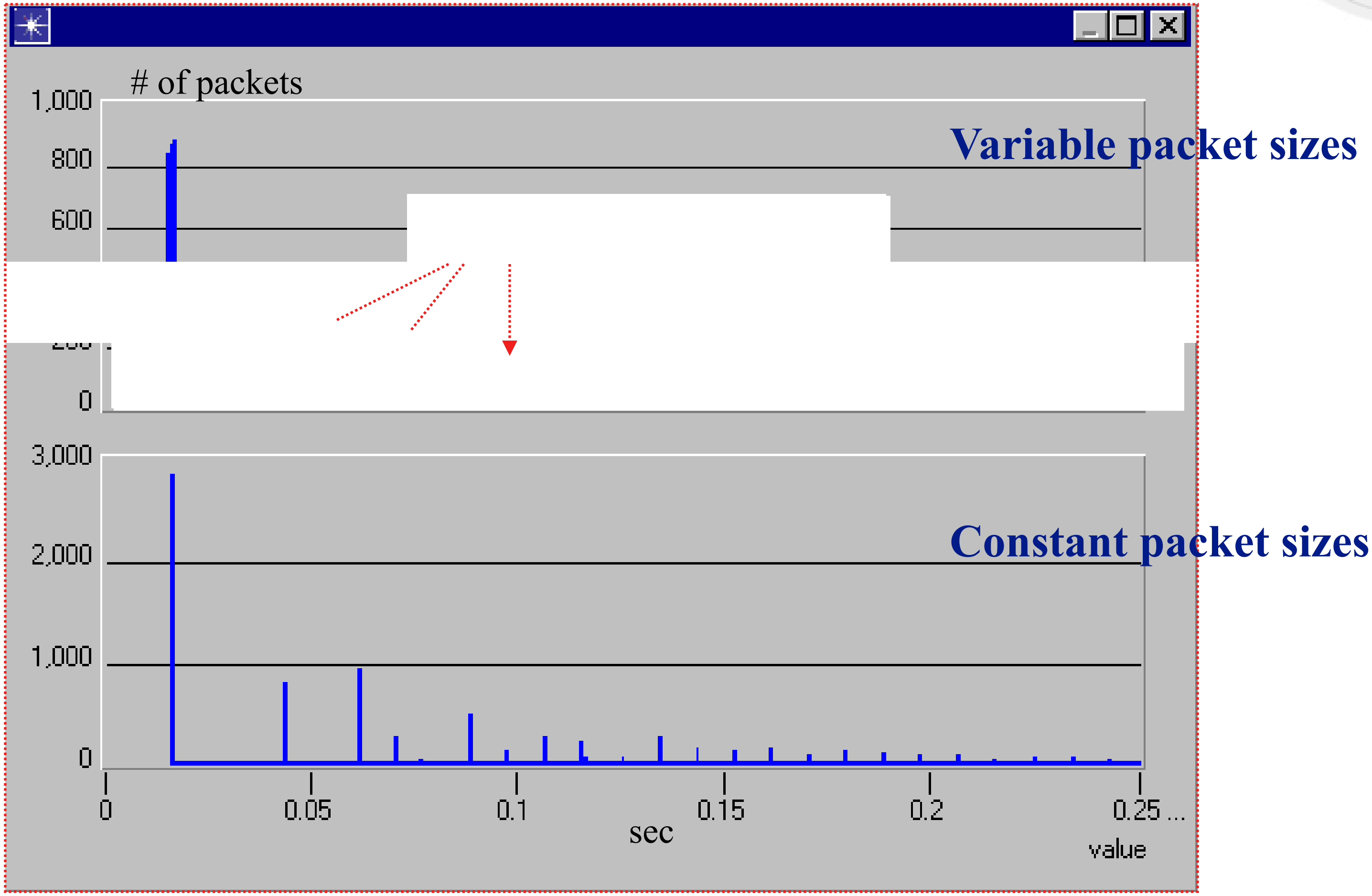
Packet Trains

Inter-departure times for small packets



Variable packet sizes

Histogram of inter-departure times for small packets



Outline

- Basic concepts
- **Source models**
  - **Poisson traffic**
  - **Batch arrivals**
-
- Single-que
- Priority/shared service systems
- Networks of queues
- Hybrid simulation (demo)

Poisson Process with Rate λ

- Interarrival times are independent and exponentially distributed
- Models well the accumulated traffic of many independent sources
- T
  (secs/pack
  (packets/sec)

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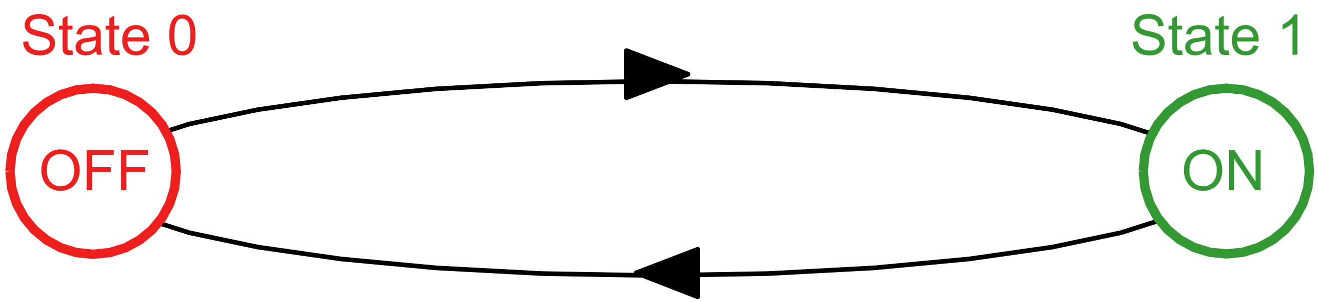
Batch Arrivals

- Some sources transmit in packet bursts
- May be better modeled by a batch arrival process (e.g., bursts of packets arriving according to a Poisson process)
- The case for a batch m eues after the first, b

M a c i n t o s h P I C T
i m a g e f o r m a t
i s n o t s u p p o r t e d

Markov Modulated Rate Process (MMRP)



distr
Tra
(e.g., Poisson, deterministic, etc) at each state

- Extension: Models with more than two states

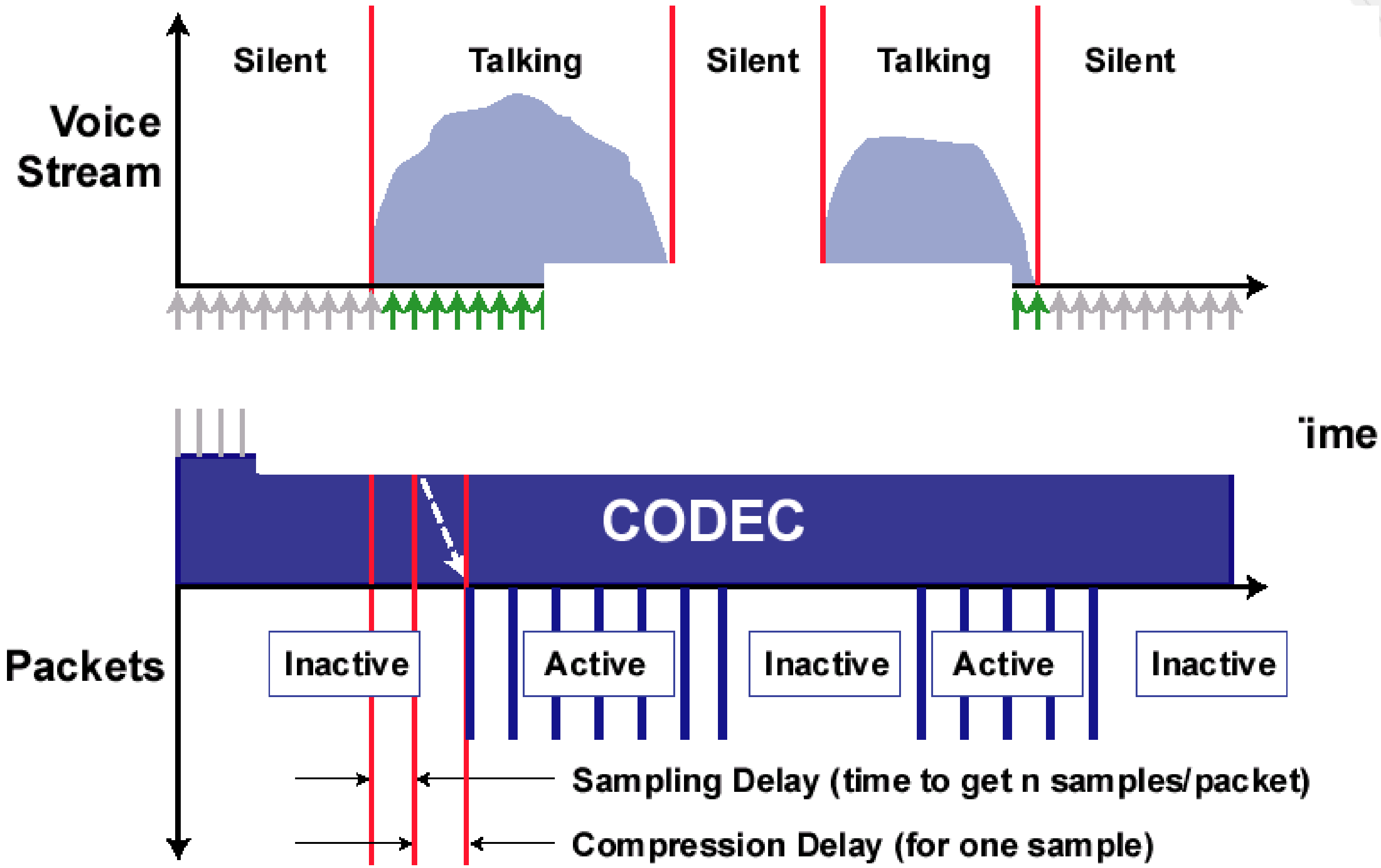
Source Types

- Voice sources
- Video sources
- File transfers
- Web traffic
- I
- Different a ents,
  e.g., delay, jitter, loss, throughput, etc.

Source Type Properties

	Characteristics	QoS Requirements	Model
Voice	* Alternating talk-spurts and silence intervals.	**Delay < ~150 ms** Jitter < ~30 ms	* Two-state (on-off) Markov Modulated Rate Process (MMRP) ponentially distributed time at
Video	* Highly bursty traffic (when encoded) * Long range dependencies	Delay < ~ 400 ms **Jitter < ~ 30 ms** Packet loss < ~1%	K-state (on-off) Markov Modulated Rate Process (MMRP)
Interactive *FTP* *telnet* *web*	* Poisson type * Sometimes batch-arrivals, or bursty, or sometimes on-off	Zero or near-sero packet loss Delay may be important	Poisson, Poisson with batch arrivals, Two-state MMRP

Typical Voice Source Behavior



MPEG1 Video Source Model

- The MPEG1 MMRP model can be extremely bursty, and has "long range dependency" behavior due to the deterministic frame sequence

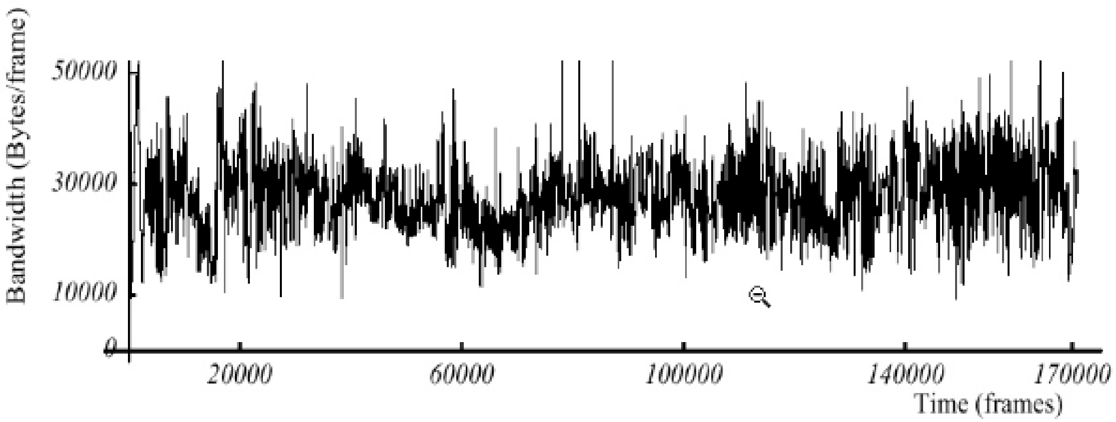


Diagram Source: Mark W. Garrett and Walter Willinger, "Analysis, Modeling, and Generation of Self-Similar VBR Video Traffic, BELLCORE, 1994

Outline

- Basic concepts
- Source models
- **Service models**
  - **Single vs. multiple-ser**
- Single-que
- Priority/shared service systems
- Networks of queues
- Hybrid simulation (demo)

Device Queuing Mechanisms

- Common queue examples for IP routers
  - FIFO: First In First Out
  - PQ: Priority Queuing
  - WFQ: Weighted Fair Queuing
-
  - Single se
  - Multiple server (one queue - several transmission lines)
  - Priority server (several queues with hard priorities - one transmission line)
  - Shared server (several queues with soft priorities - one transmission line)

Single Server FIFO

- Single transmission line serving packets on a FIFO (First-In-First-Out) basis
- Each packet must wait for all packets found in the system to complete transmission smission

- Packets ar

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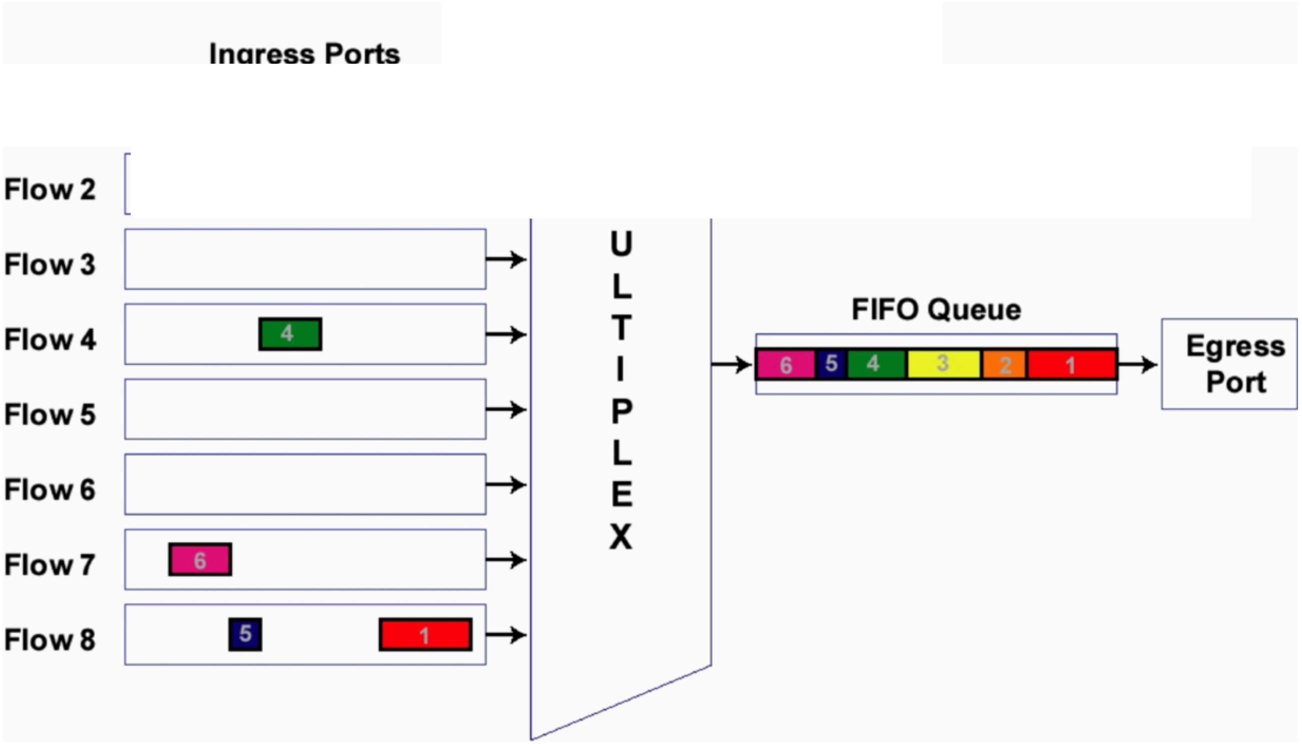
FIFO Queue

- Packets are placed on outbound link to egress device in FIFO order
  - Device (router, switch) multiplexes different flows arriving on various ingress ports onto an output buffer forming a FIFO queue



Multiple Servers

- Multiple packets are transmitted simultaneously on multiple lines/servers
- Head of the line service: packets wait in a FIFO queue, and when a server become t goes into service

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Priority Servers

- Packets form priority classes (each may have several flows)
- There is a separate FIFO queue for each priority class
- Packets of lower priority start transmission only if no higher priority packet is waiti
-

  packet fo
  - Preemptive (high priority packet does not have to wait …)

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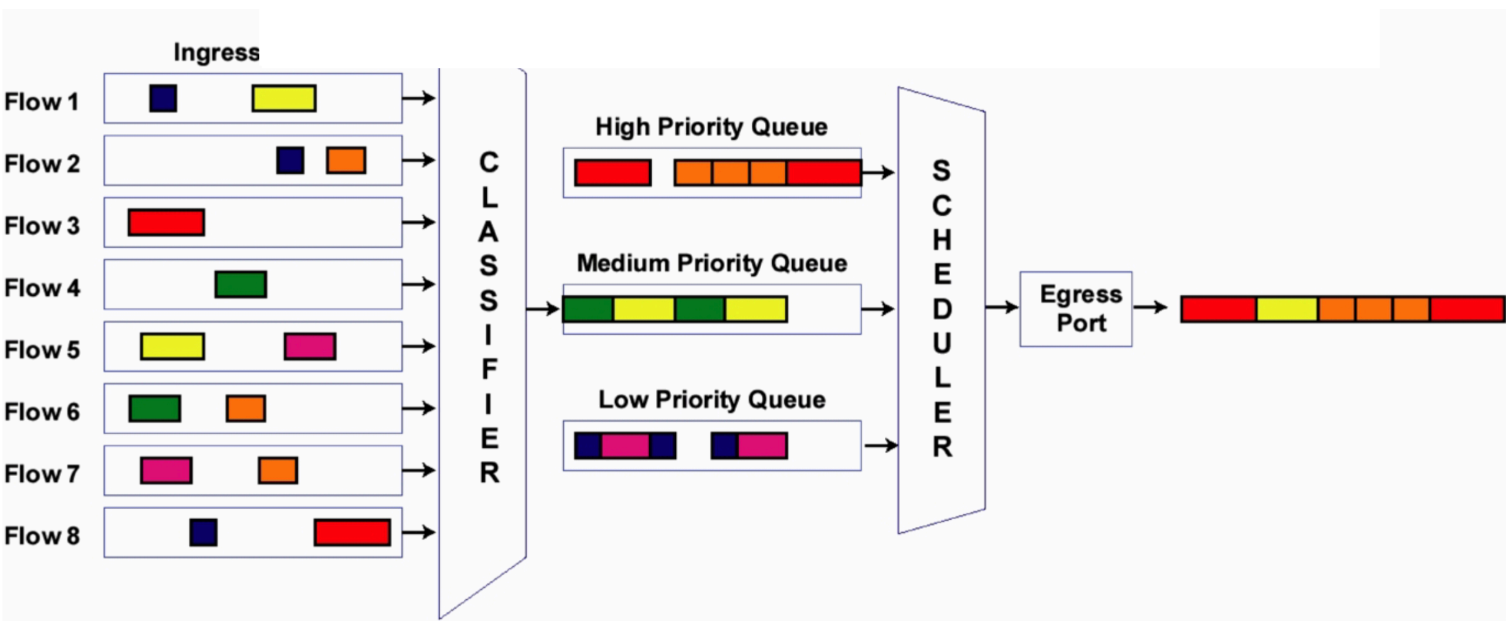
Priority Queuing

- Packets are classified into separate queues
  - E.g., based on source/destination IP address, source/destination TCP port, etc.
- All packets in a higher priority queue are served before a lower priority queue is served
  - Typically in routers, if a hi while a lower priority



Shared Servers

- Again we have multiple classes/queues, but they are served with a "soft" priority scheme
- Round-robin
- Weighted fair queuing

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Round-Robin/Cyclic Service

- Round-robin serves each queue in sequence
  - A queue that is empty is skipped
  - Each queue when served may have limited service (at most k packets transmitted with k = 1 or k > 1)
-

  n
- Round-rob ation

  among the queues.

Fair Queuing

- This scheduling method is inspired by the "most fair" of methods:
  - Transmit one bit from each queue in cyclic order (bit-by-bit round robin)
  - Skip queues that are empty
- To approximate the bit-by-bit processing behavior, for each packet
  - We calculate upon arrival its "finish time under bit-by-bit round robin" assuming all other queues we transmit by FIFO
- Important pr
  - Priority is given to short packets
  - Equal bandwidth is allocated to all queues that are continuously busy

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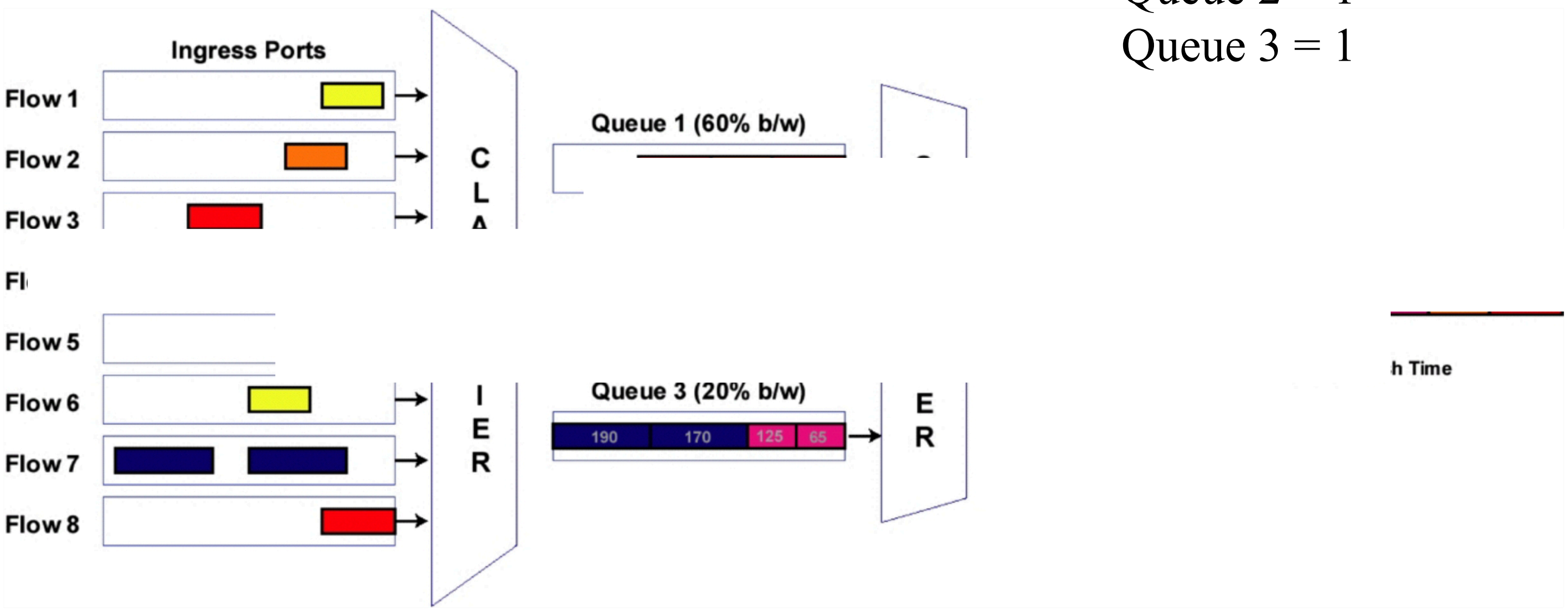
Weighted Fair Queuing

- Fair queuing cannot be used to implement bandwidth allocation and soft priorities
- Weighted fair queuing is a variation that corrects this deficiency
  - Let w_k be the weight of the k^{th} queue
  - Think of round-robin with ts upon its turn
  - If all queues have always s ueue receives bandwidth
- F
- Priority queu move to higher priorities
- Again, to deal with the segmentation problem, we approximate as follows: For each packet:
  - We calculate its "finish time" (under the weighted bit-by-bit round robin scheme)
  - We next transmit the packet with the minimum finish time

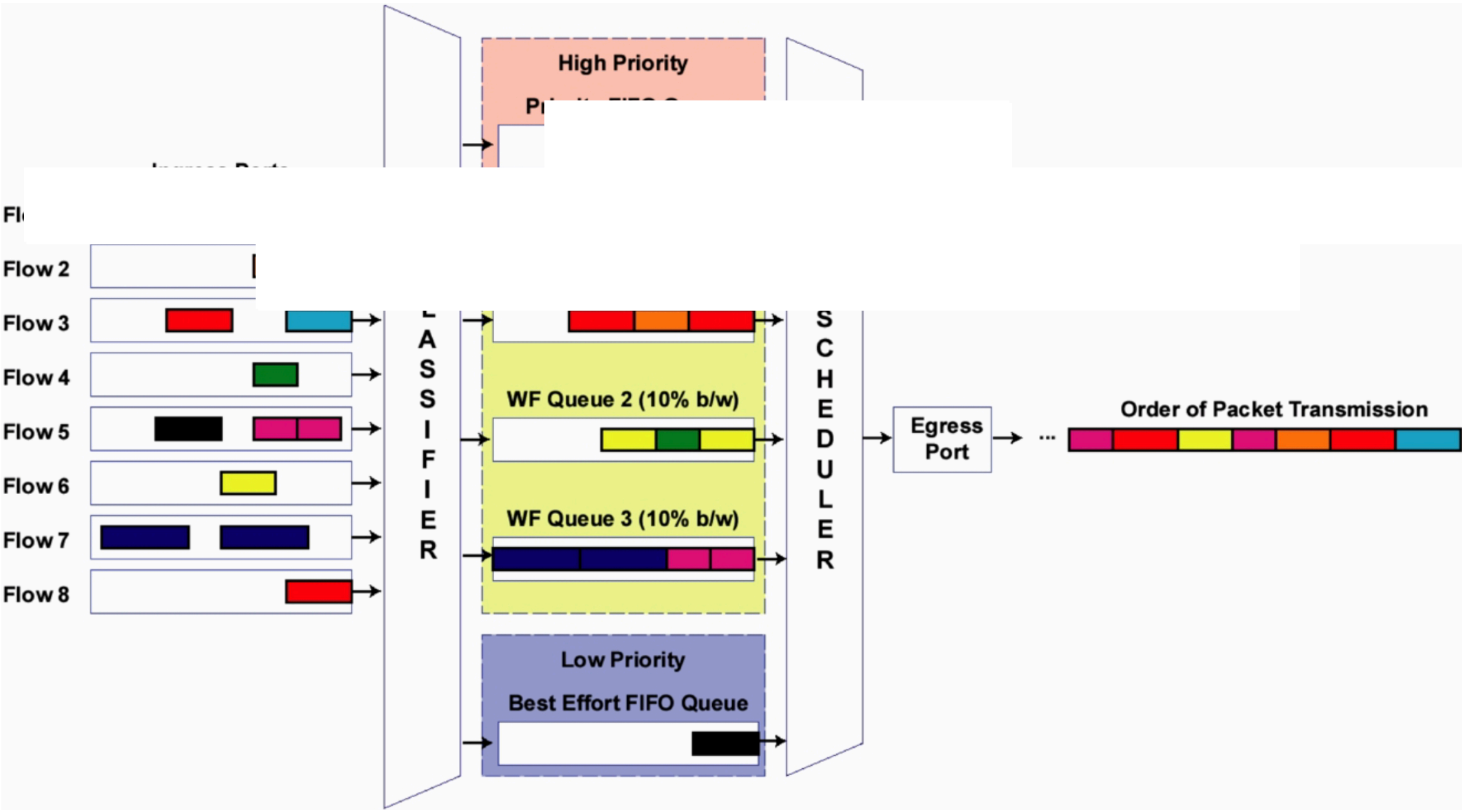
Weighted Fair Queuing Illustration

Weights:
Queue 1 = 3
Queue 2 = 1
Queue 3 = 1

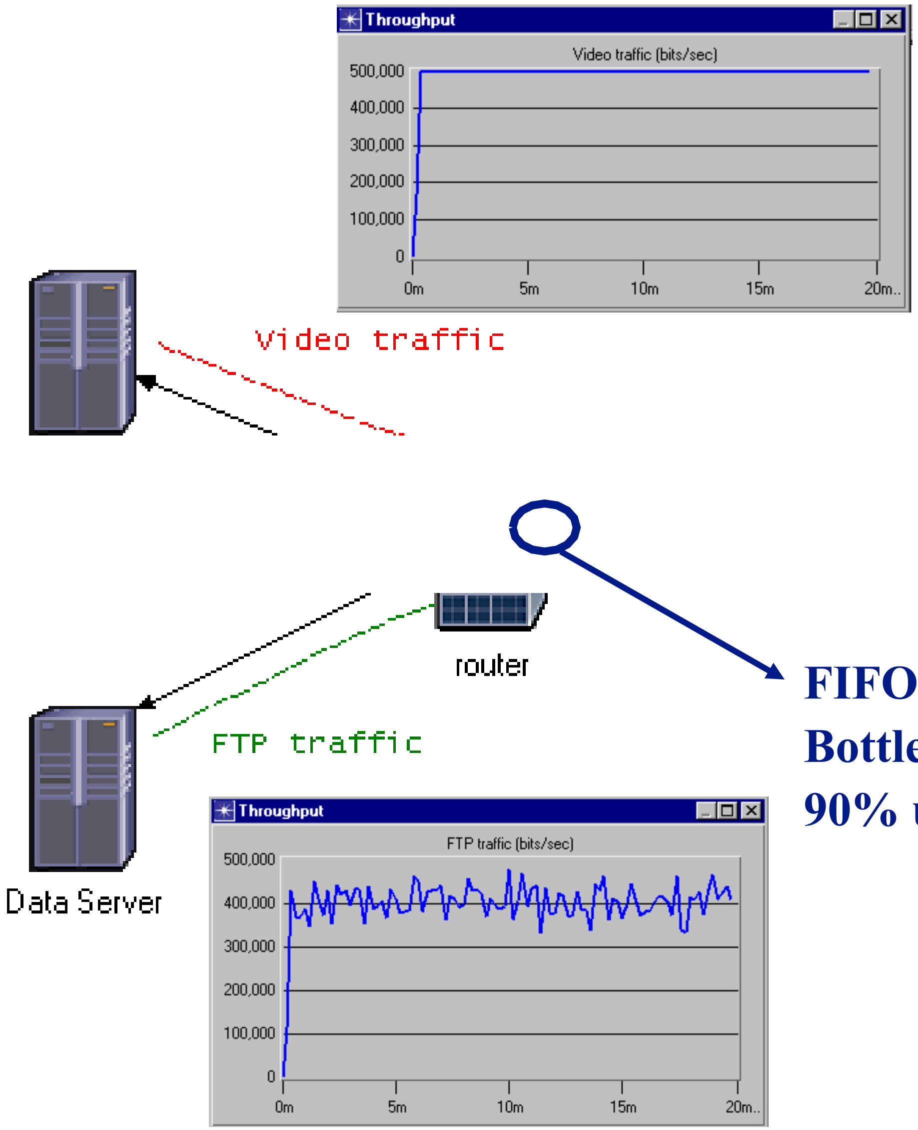


Combination of Several Queuing Schemes

- Example – voice (PQ), guaranteed b/w (WFQ), Best Effort (Cisco's LLQ implementation)

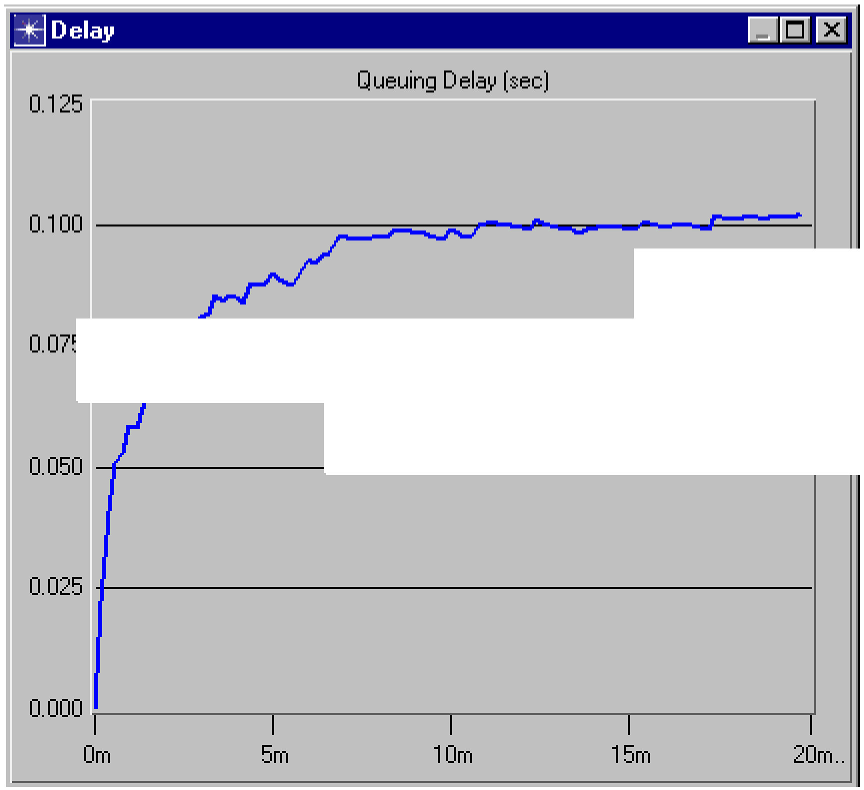


Demo: FIFO



FIFO
Bottleneck
90% utilization

Demo: FIFO Queuing Delay



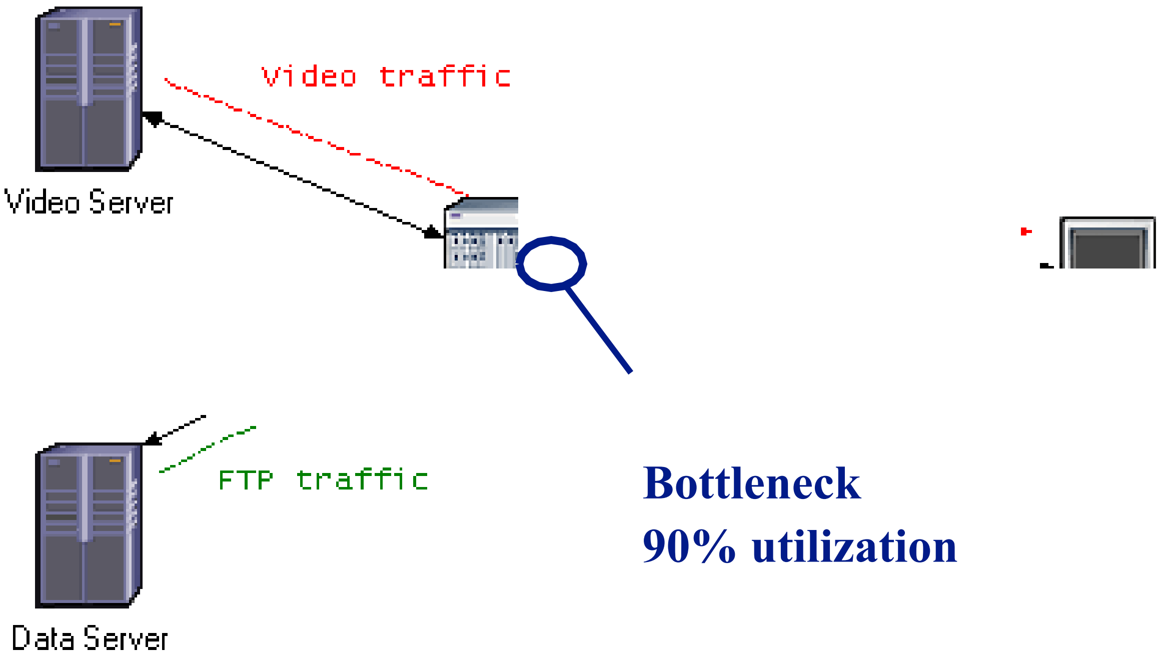
Applications have different requirements

- Video
  - » delay, jitter

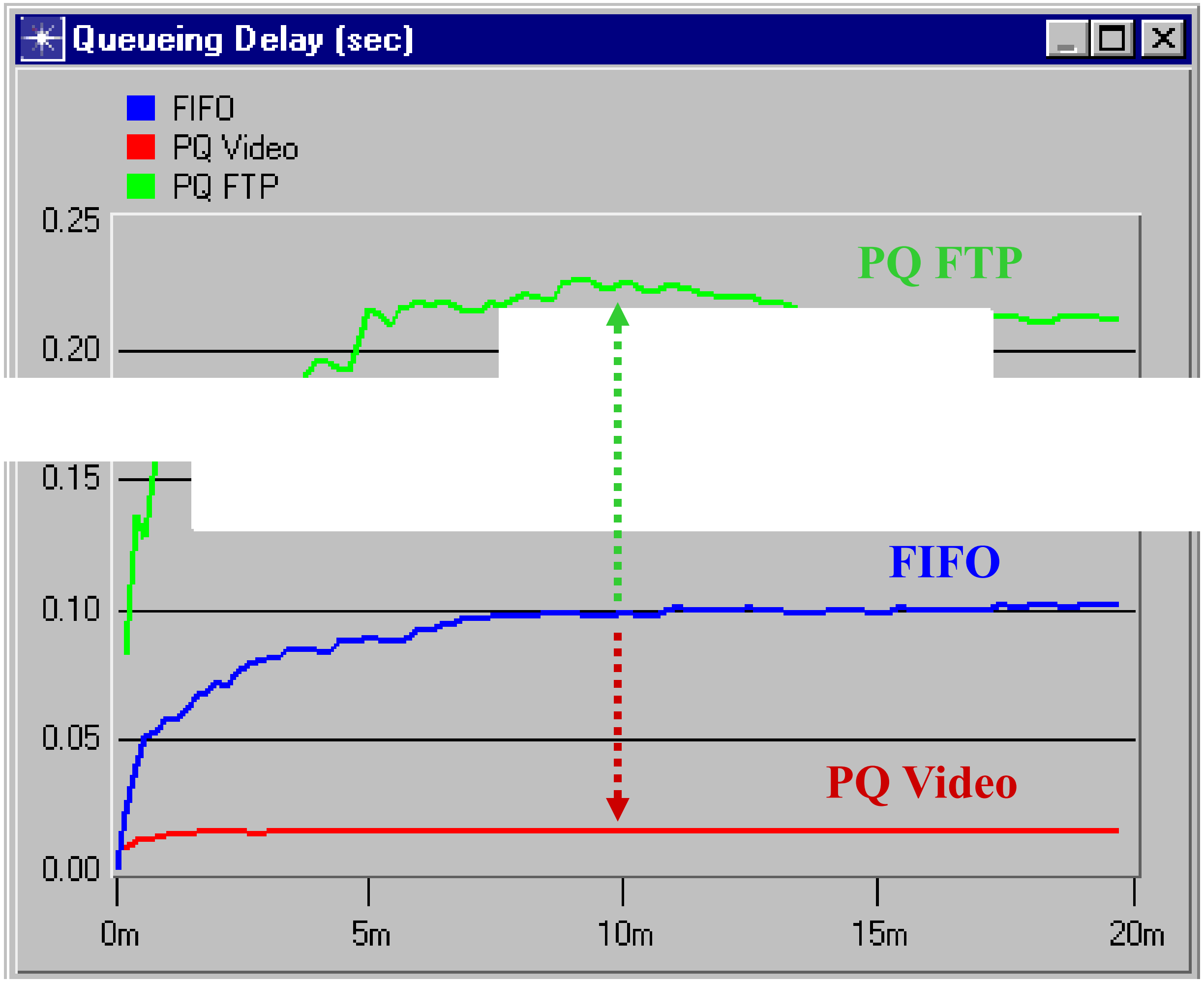
rt” needed

- Priority Queuing (PQ)
- Weighted Fair Queuing (WFQ)

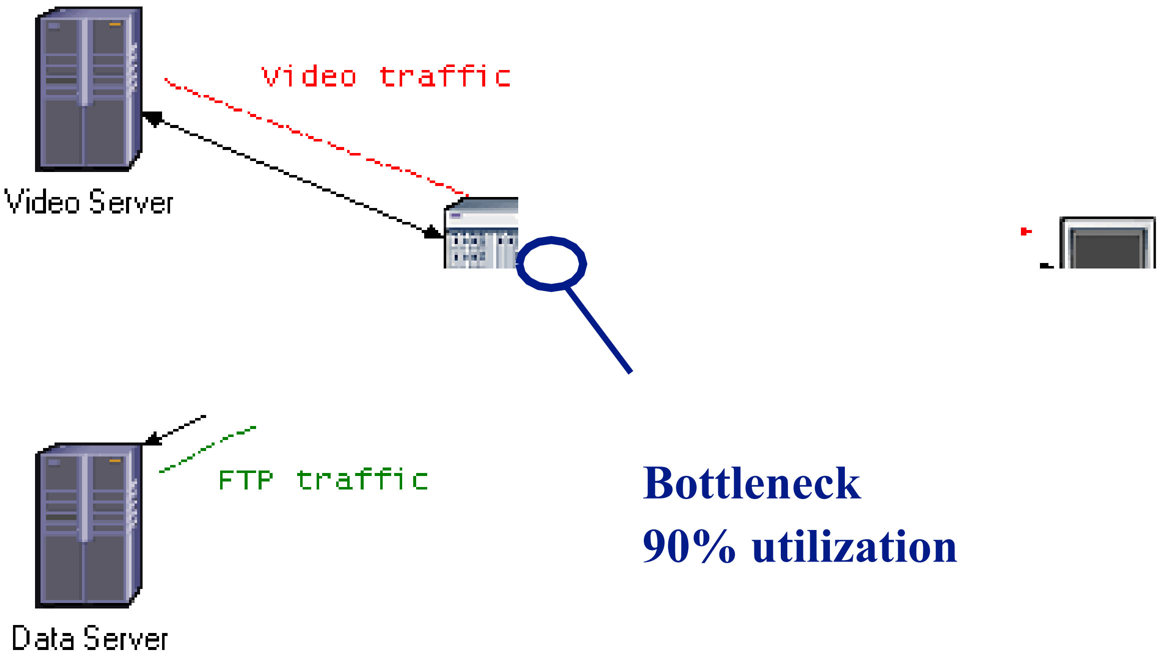
Demo: Priority Queuing (PQ)



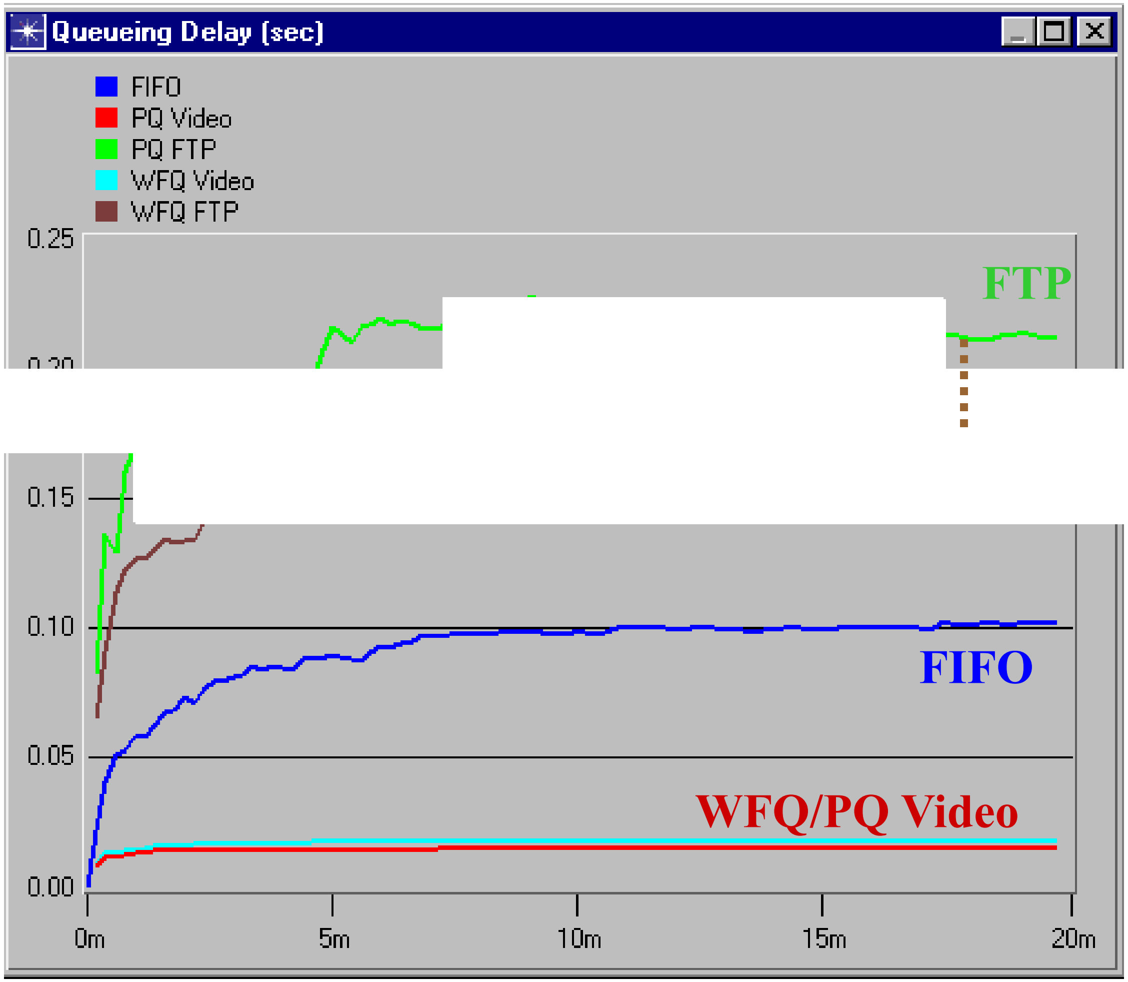
Demo: PQ Queuing Delays



Demo: Weighted Fair Queuing (WFQ)



Demo: WFQ Queuing Delays



Queuing: Take Away Points

- Choice of queuing mechanism can have a profound effect on performance
- To achieve desired service differentiation, appropriate queuing mechanisms can be used
- Complex queuing mec simulation t
- Improper tion or weights) may impact performance of low priority traffic

Outline

- Basic concepts
- Source models
- Service models (demo)
- **Single-queue systems**
  - **Demo: Analytics vs. simulation**
- Priority/shared service systems
- Networks of queues
- Hybrid simulation (demo)

M/M/1 System

- Nomenclature: M stands for "Memoryless" (a property of the exponential distribution)
  - **M**/M/1 stands for Poisson arrival process (which is memoryless)
  - M/**M**/1 stands for exponentially distributed transmission times
-
  - Packet tra ean $1/\mu$
  - One server
  - Independent interarrival times and packet transmission times
- Transmission time is proportional to packet length
- Note $1/\mu$ is secs/packet so μ is packets/sec (packet transmission rate of the queue)
- Utilization factor: $\rho = \lambda/\mu$ (stable system if ρ ■ 1)

Delay Calculation

- Let

  Q = Average time spent waiting in queue

  T = Average packet delay (transmission plus queuing)

- Note that $T = 1/\mu + Q$
-

  N =

where

  N_q = Average number waiting in queue

- These quantities can be calculated with formulas derived by Markov chain analysis (see references)

M/M/1 Results

- The analysis gives the steady-state probabilities of number of packets in queue or transmission
- $P\{n \text{ packets}\} = \rho^n(1-\rho)$ where $\rho = \lambda/\mu$
-

$T =$

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Example: How Delay Scales with Bandwidth

- Occupancy and delay formulas

  $N = \rho/(1 - \rho)$ $T = 1/(\mu - \lambda)$ $\rho = \lambda/\mu$

- Assume:

  – System tr

- Then:
  – Queue sizes stay at the same level (ρ stays the same)
  – Packet delay is cut in half (μ and λ are doubled)
- A conclusion: In high speed networks
  – propagation delay increases in importance relative to delay
  – buffer size and packet loss may still be a problem

M/M/m, M/M/□ System

- Same as M/M/1, but it has m (or □) servers
- In M/M/m, the packet at the head of the queue moves to service when a server becomes free
-
- There are
  probabilities and average delay of these systems

Finite Buffer Systems: M/M/m/k

- The M/M/m/k system
  - Same as M/M/m, but there is buffer space for at most k packets. Packets arriving at a full buffer are dropped
- Formulas for avera   tate occupancy
- The M/

  telephone or circuit switching systems

Characteristics of M/M/. Systems

- Advantage: Simple analytical formulas
- Disadvantages:
  - The Poisson assumption may be violated
  -
  - Interarr
    dependent (particularly in the network core)
  - Head-of-the-line assumption precludes heterogeneous input traffic with priorities (hard or soft)

M/G/1 System

- Same as M/M/1 but the packet transmission time distribution is general, with given mean $1/\mu$ and variance σ^2
- Utilization factor
-

Average t

Average delay = $1/\mu + \lambda(\sigma^2 + 1/\mu^2)/2(1- \times)$

- The formulas for the steady-state occupancy probabilities are more complicated
- Insight: As σ^2 increases, delay increases

G/G/1 System

- Same as M/G/1 but now the packet interarrival time distribution is also general, with mean ✦ and variance ■²
- We still assume FIF   nt interarrival t
- Heavy tr

  Average time in queue ~ $\lambda(\sigma^2 + $ ■$^2)/2(1-$ ✕ $)$
- Becomes increasingly accurate as ✕ 🎭 □

Demo: M/G/1

Packet inter-arrival times
exponential (0.02) sec

Capacity
1 Mbps

Packet size
1250 bytes
(10000 bits)

Packet size distribution:
exponential
constant
lognormal

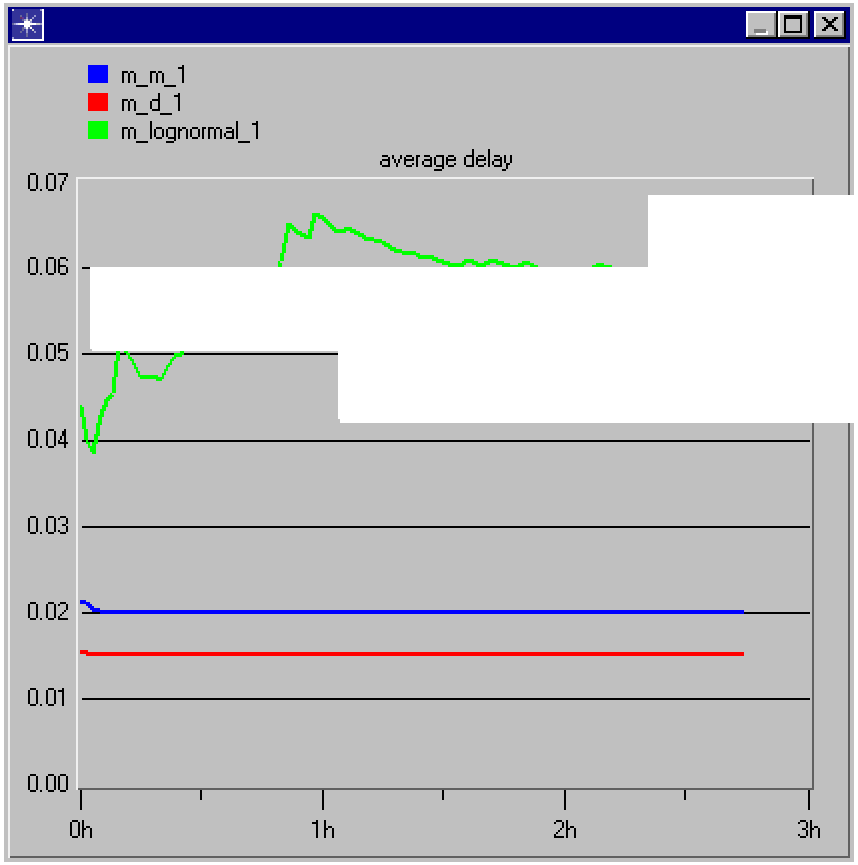
What is the average delay and queue size ?

Demo: M/G/1 Analytical Results

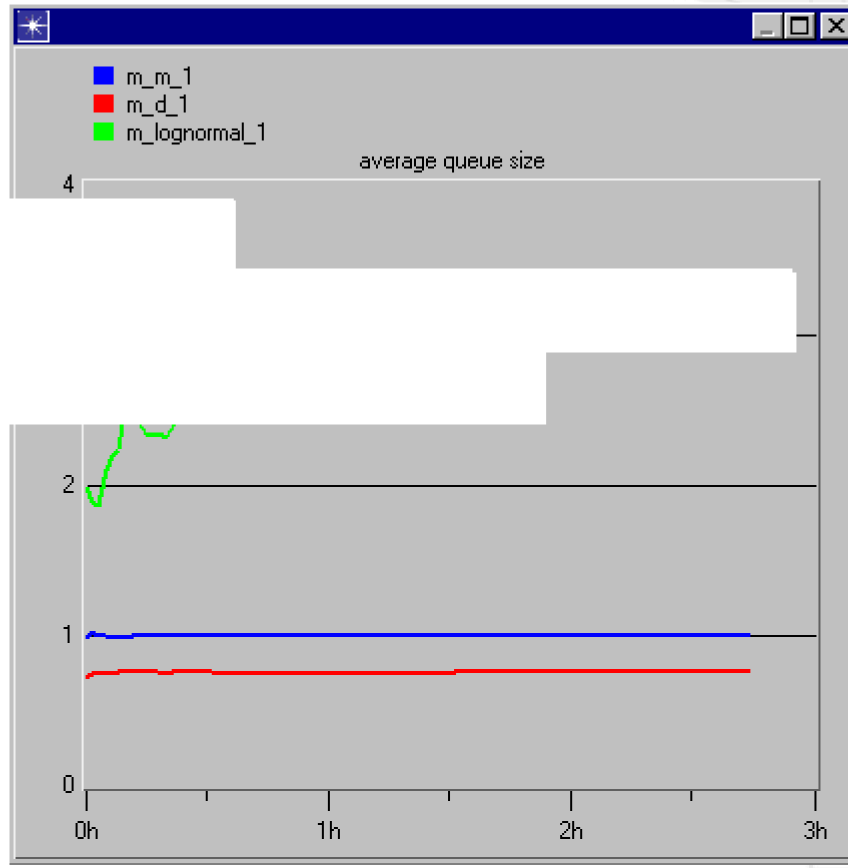
Packet Size Distribution	Delay T (sec)	Queue Size (packets)
Exponential		
Consta $mean = 10000$ $variance = \text{N/A}$	0.015	0.75
Lognormal $mean = 10000$ $variance = 9.0 * 10^8$	0.06	3.0

Demo: M/G/1 Simulation Results

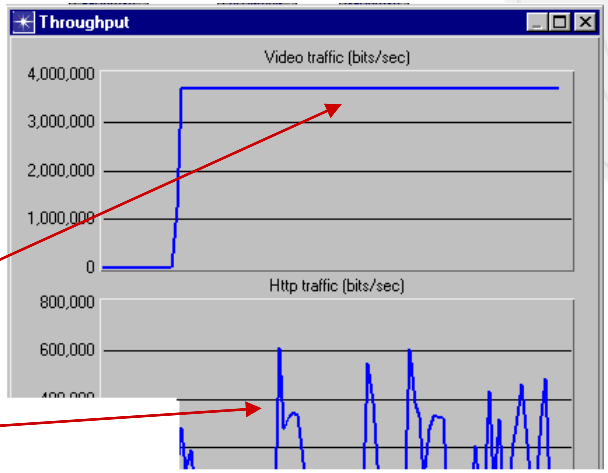
Average Delay (sec)



Average Queue Size (packets)

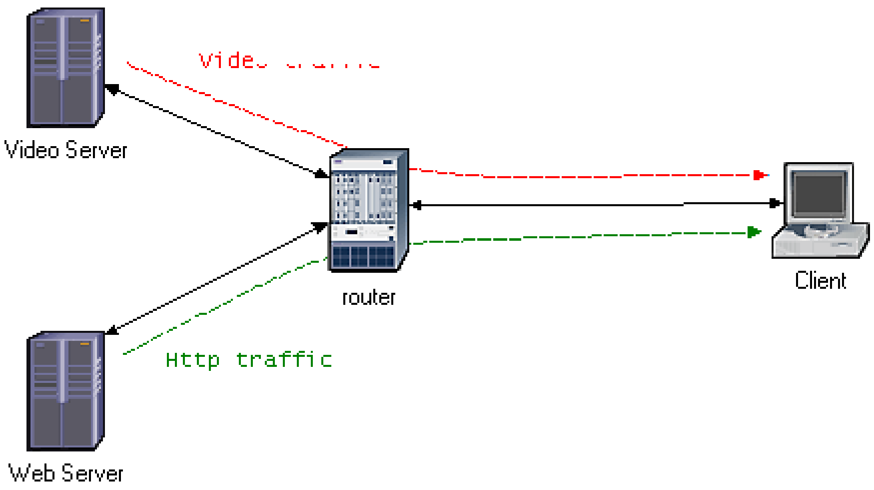


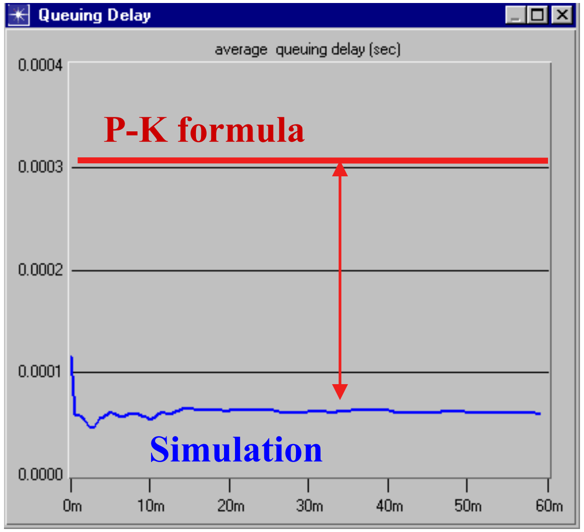
Demo: M/G/1 Limitations



Application traffic mix not memoryless

- **Video**
  - » constant packet inter-arrivals
- **Http**





Outline

- Basic concepts
- Source models
- Service models (demo)
- Single-queue systems
-
  - **Preempt**
  - **Cyclic, WFQ, PQ systems**
  - **Demo: Simulation results**
- Networks of queues
- Hybrid simulation (demo)

Non-preemptive Priority Systems

- We distinguish between different classes of traffic (flows)
- Non-preemptive priority: packet under transmission is not preempted by a packet of higher priority
- P-K formula for delay

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Cyclic Service Systems

- Multiple flows, each with its own queue
- Fair system: Each flow gets access to the transmission line in turn
- Several possible assu any packets each f
- available

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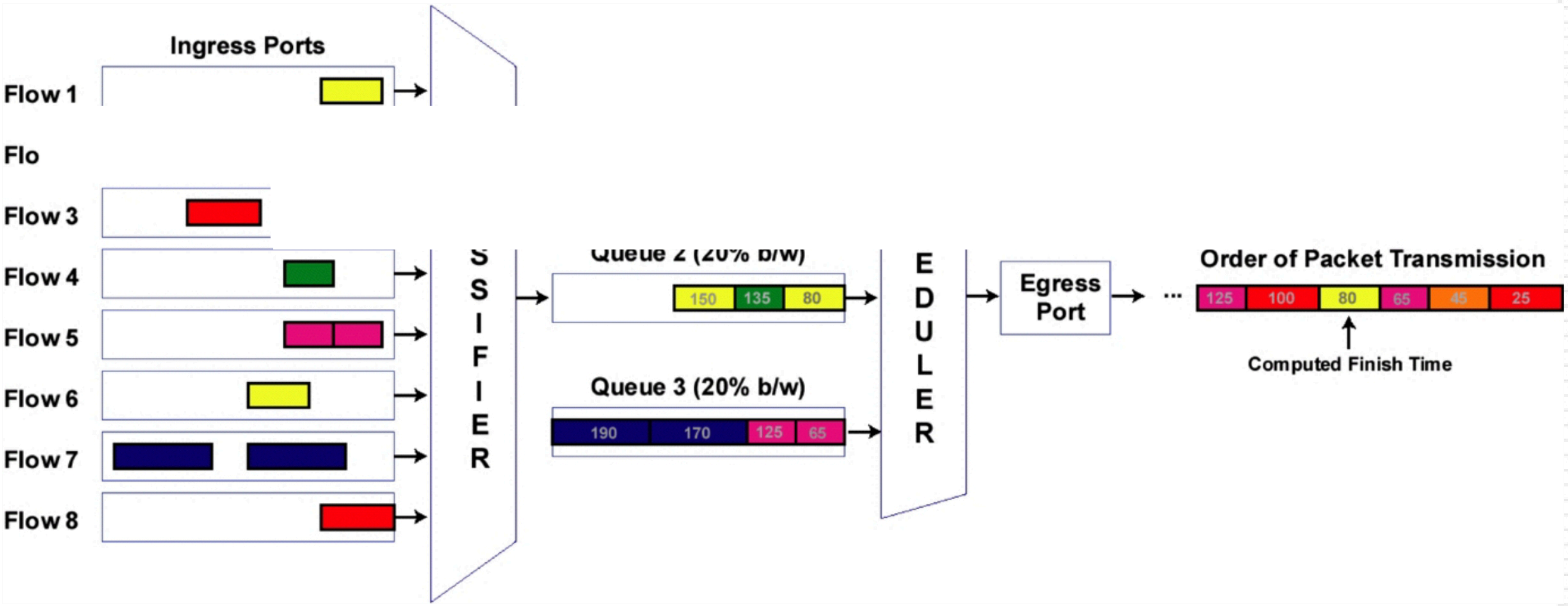
Weighted Fair Queuing

- A combination of priority and cyclic service
- No exact analytical formulas are available



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- Basic concepts
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- Single-queue systems
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- **Networks**
  - **Violation of M/M/. assumptions**
  - **Effects on delays and traffic shaping**
  - **Analytical approximations**
- Hybrid simulation (demo)

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Two Queues in Series

- First queue shapes the traffic into second queue
- Arrival times and packet lengths are correlated
- M/M/1 and M/G/1 formulas yield significant error for second queue

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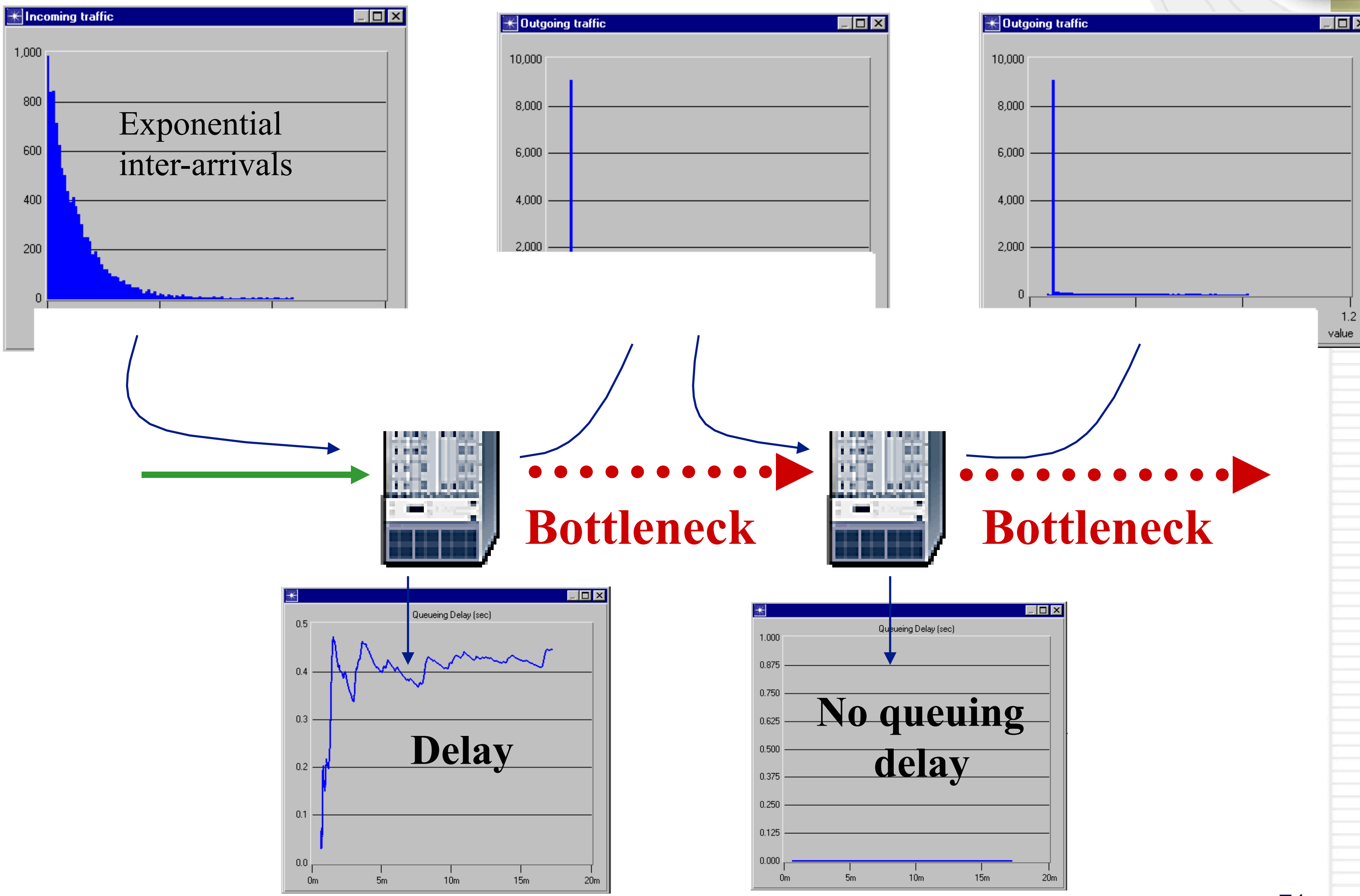
Two bottlenecks in series



Approximations

- Kleinrock independence approximation
  - Perform a delay calculation in each queue independently of other queues
  - Add the results (including propagation delay)
-

  a
- Tends to b ffic mixing", e.g., nodes serving many relatively small flows from several different locations

Outline

- Basic concepts
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- Single-queue systems
-
- Networks
- **Hybrid simulation**
  - **Explicit vs. aggregated traffic**
  - **Conceptual Framework**
  - **Demo: PQ and WFQ with aggregated traffic**

Basic Concepts of Hybrid Simulation

- Aims to combine the best of analytical results and simulation
- Achieve significant gain in simulation speed with little loss of accuracy
- Divides the traffic thro **licit** and
  - Backgrou
- The interaction of explicit and background is modeled either analytically or through a "fast" simulation (or a combination)

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Explicit Traffic

- Modeled in detail, including the effects of various protocols
- Each packet's arrival and departure times are recorded (together with other data of interest, e.g., loss, etc.) along each link that it traverses
- Departure times at a link t the next link (plus
-
  lengths), d

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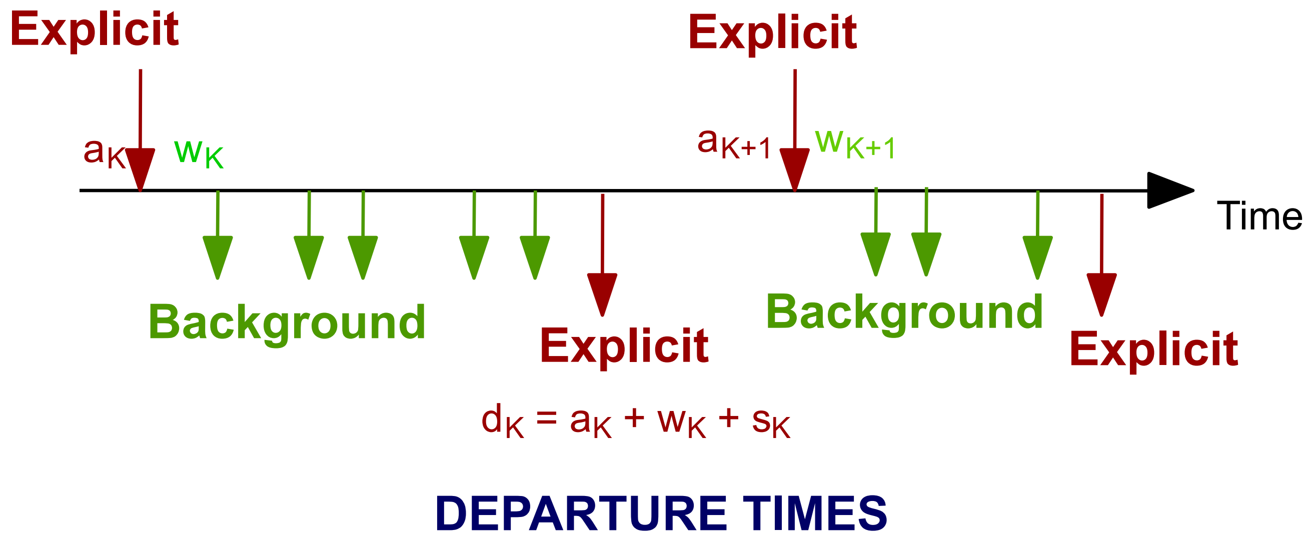
Aggregated Traffic

- Simplified modeling
  - We don't keep track of individual packets, only workload counts (number of packets or bytes)
  - We "generate" workload counts
    - » by probabilistic/analyti

- Aggregate
- Shaping effects are complex to incorporate
- Some dependences between explicit and background traffic along a chain of links are complicated and are ignored

Hybrid Simulation (FIFO Links): Conceptual Framework

- Given the arrival time a_k of the k^{th} explicit packet
- Generate the workload w_k found in queue by the k^{th} packet
- From a_k and w_k generate the departure time of the k^{th} packet as



Simulating the Background Traffic Effects

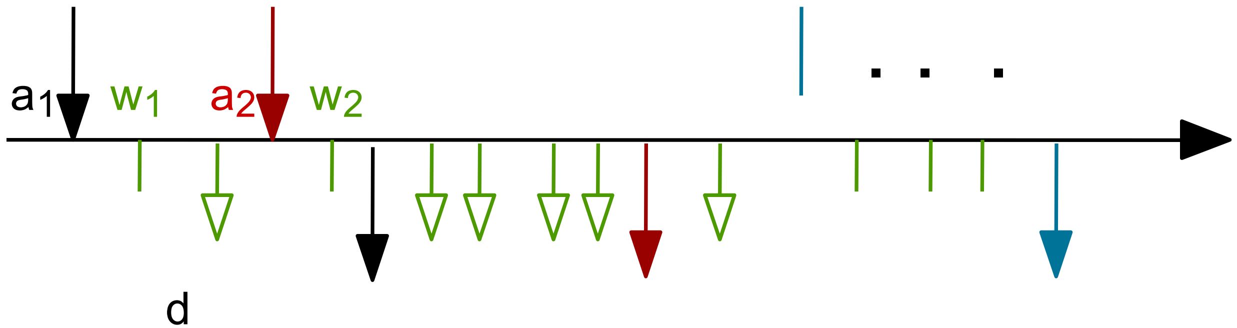
- Use a traffic descriptor for the background traffic (e.g., carried by special packets)
- Traffic descriptor includes:
  - Traffic volume information (e.g., packets/sec, bits/sec)
  - Probability distribution

- Generate k ns thereof
  - Successive sampling (for FIFO case)
  - Steady-state queue length distribution (if we can get it)
  - Simplified simulation (microsim - applies to complex queuing disciplines)

Hybrid Simulation (FIFO Case)

- Critical Question: Given arrival times a_k and a_{k+1}, workload w_k, and background traffic descriptor, how do we find w_{k+1}?

Arrival times/Workload found

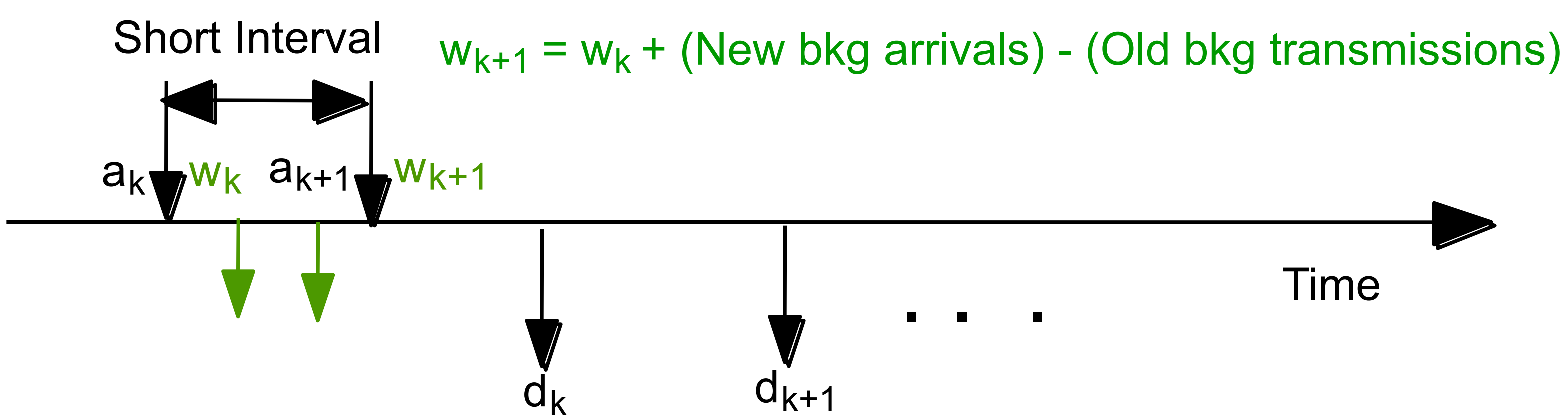


Departure times

- Note: w_{k+1} consists of w_k and two more terms:
  - Background arrivals in interval a_{k+1} - a_k
  - (Minus) transmitted workload in interval a_{k+1} - a_k
- Must calculate/simulate the two terms
- The first term is simulated based on the traffic descriptor of the background traffic
- The second term is easily calculated if the queue is continuously busy in a_{k+1} - a_k

Short Interval Case (Easy Case)

- Short interval a_{k+1} - a_k (i.e., $a_{k+1} < d_k$)
- Queue is busy continuously in a_{k+1} - a_k
- So w_{k+1} is quickly simulated
  - Sample the background on to simulate the new
  - Do the ac kload in a_{k+1} - a_k)



Long Interval Case

- Long interval a_{k+1} - a_k (i.e., $a_{k+1} > d_k$)
- Queue may be idle during portions of the interval a_{k+1} - a_k
- Need to generate/simulate
  - The new arrivals in a_{k+1} - a_k
  - The lengths of the busy pe
-

  p
- Other alterna

M a c i n t o s h P I C T
i m a g e f o r m a t
i s n o t s u p p o r t e d

Steady-State Queue Length Distribution

- If the interval between two successive explicit packets is very long, we can assume that the queue found by the second packet is in steady state
- So, we can obtain w_{k+1} ady-state d
- found or c
  - M/M/1 and other M/M/. Queues
  - Some M/G/. systems

Micro Simulation: Conceptual Framework

- Handles complex queuing systems
  - Micro-packets are generated to represent traffic load within the context of the queue only (i.e., they are not transmitted to any external links)
  - For long intervals, wher y-state is likely
    - » Sampl
- Microsim speeds up the simulation without sacrificing accuracy
- Microsim provides a general framework
  - Applies to non-stationary background traffic
  - Applies to non-FIFO service models (with proper modification)

Examples of Applications

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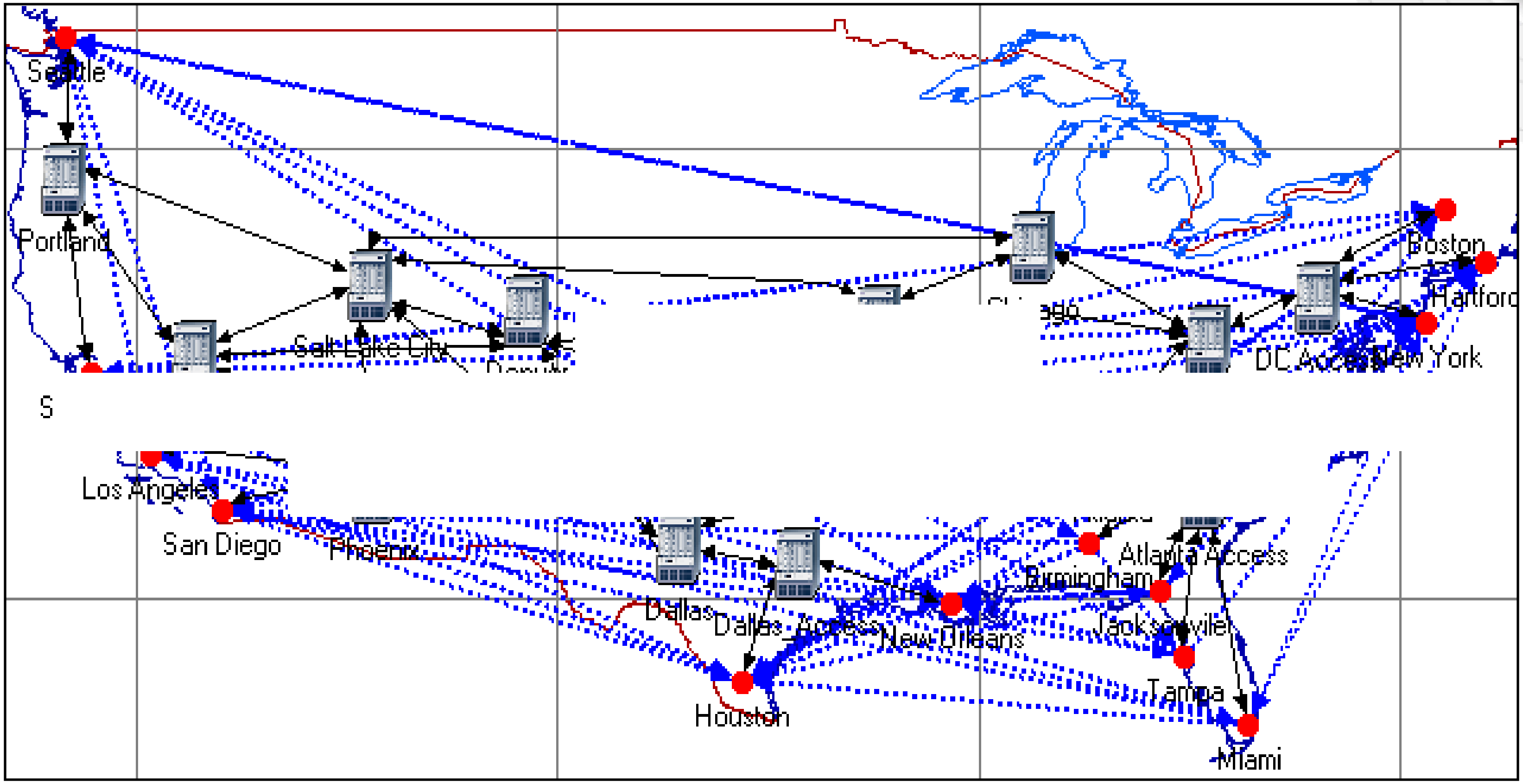
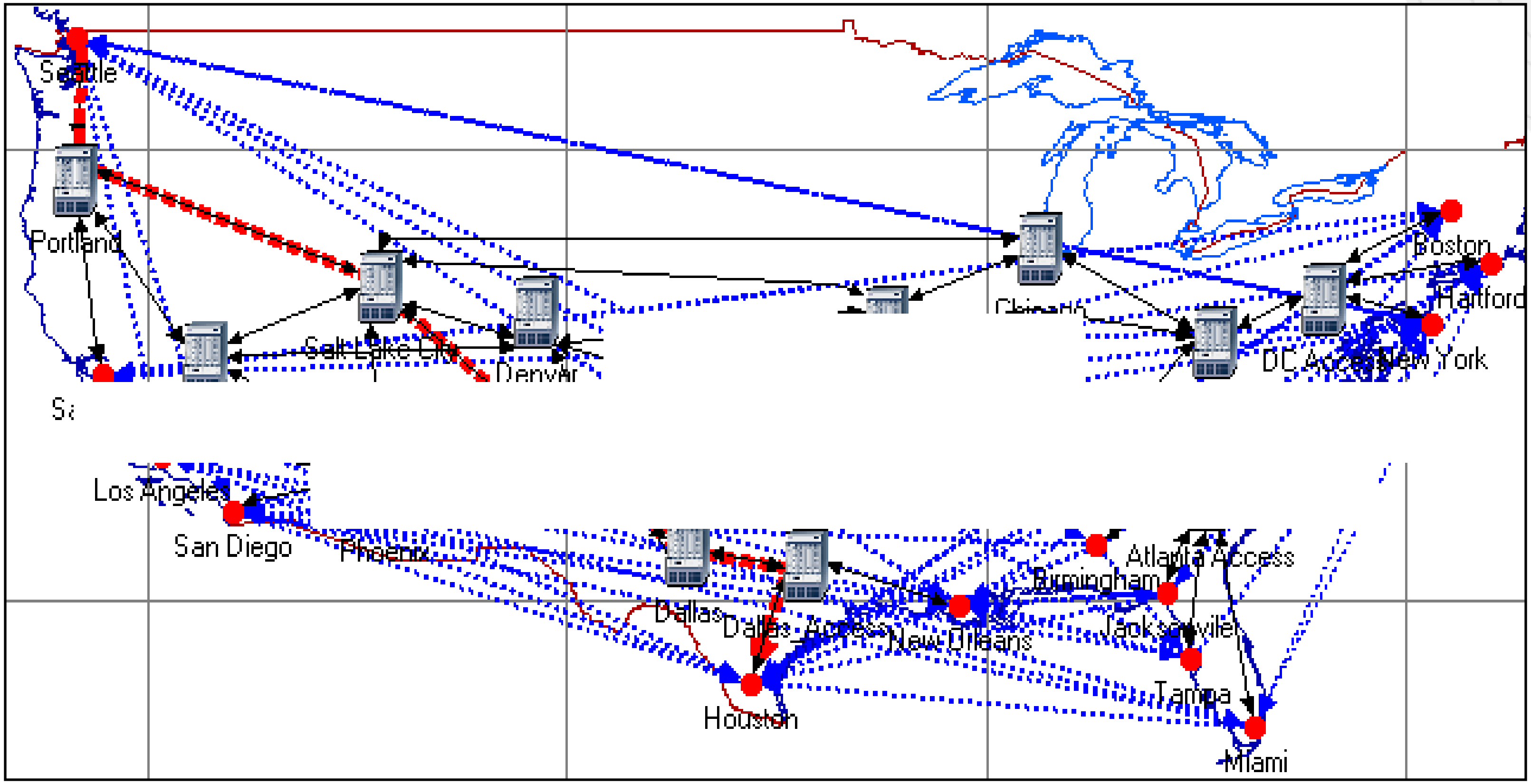
Demo End-to-end Delay: Baseline Network



Traffic modeled as

1) Explicit traffic
2) Background traffic

Target Flow: ETE delay as a function of ToS

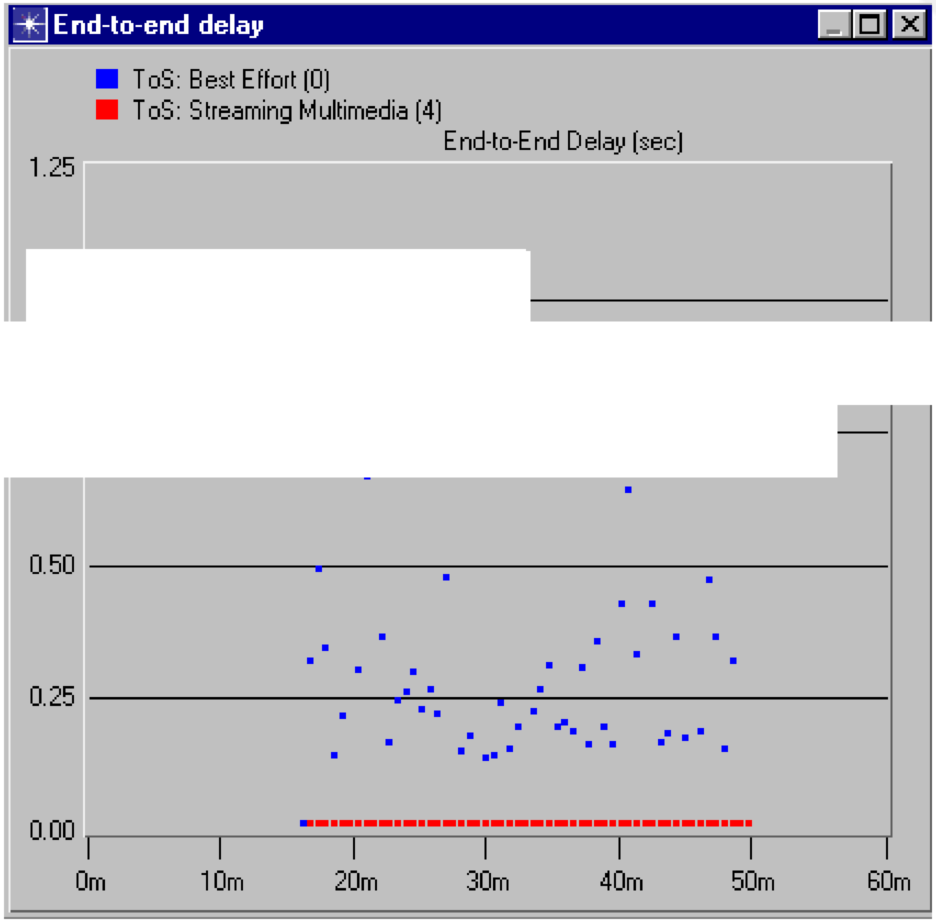


Target flow: Seattle → Houston - modeled using explicit traffic

- Varying its Type of Service (ToS)
  - » Best Effort (0)
  - » Streaming Multimedia (4)

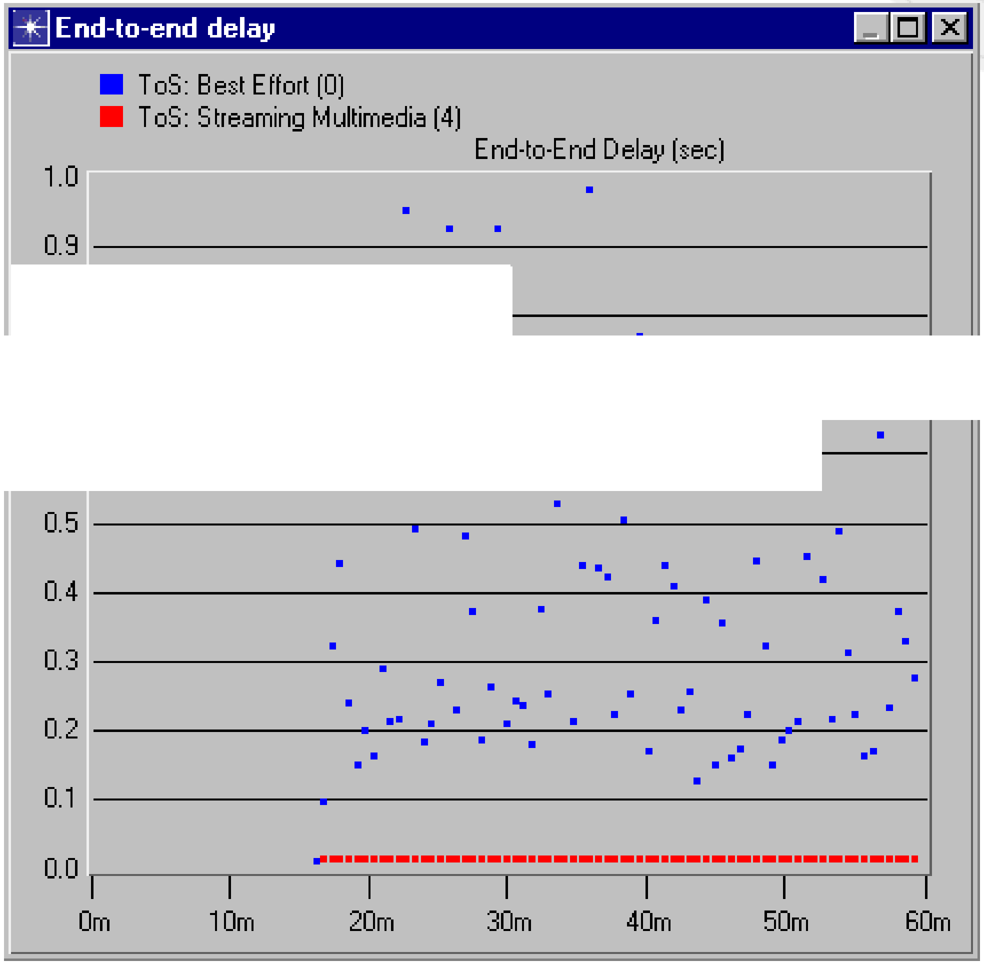
Explicit Simulation Results for Target Flow

- Total traffic volume
  - » 500 Mbps
- Time modeled
- Simulatio
  - » **31 hours**

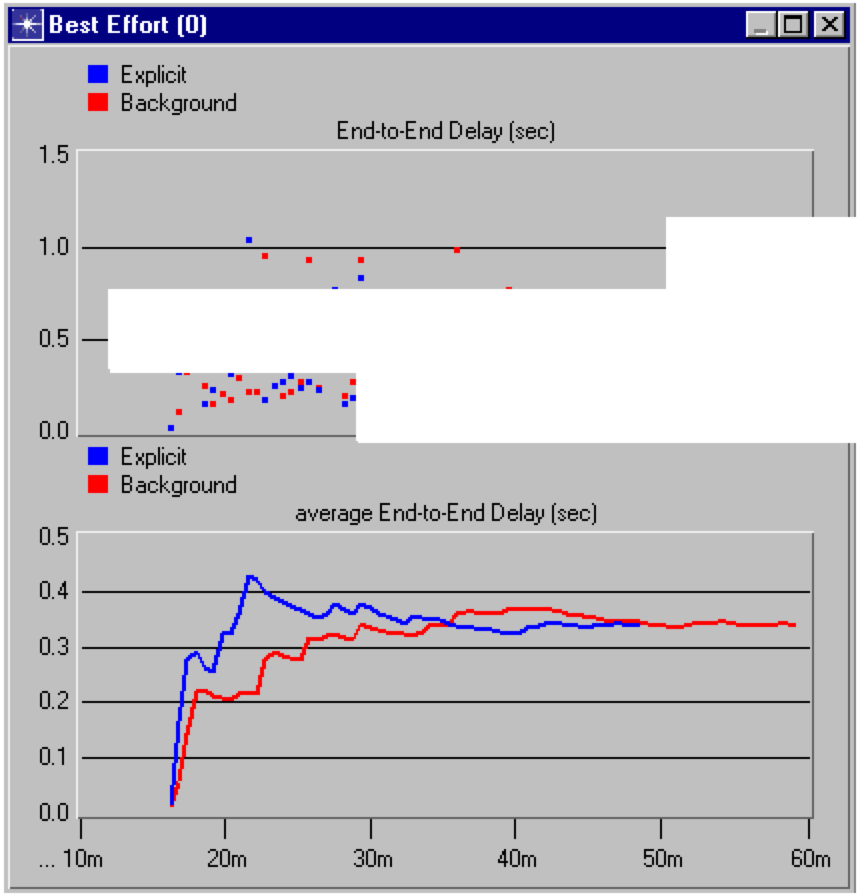


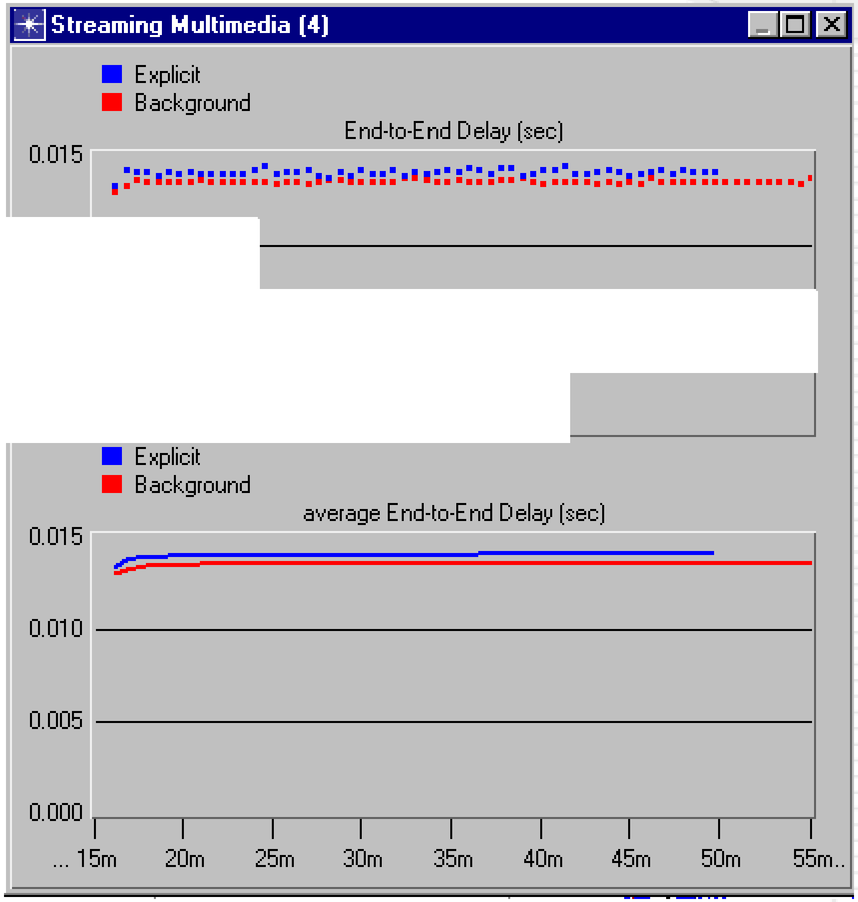
Hybrid Simulation Results for Target Flow

- Total traffic volume
  - » 500 Mbps
- Time modeled
- Simulatio
  - » **14 minutes**



Comparison: Hybrid vs Explicit Simulation





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